

IMPERIAL

HIGH-DIMENSIONAL FINANCIAL DATA ANALYSIS WITH TENSOR ATTENTION-BASED NEURAL NETWORKS

Author

IOANNIS P. ZIOULIS

CID: 02223077

Supervised by

PROF. DANILO MANDIC

Second Marker

Prof. Cong Ling

A Thesis submitted in fulfillment of requirements for the degree of
Master of Engineering in Electronic and Information Engineering

Department of Electrical and Electronic Engineering
Imperial College London
2026

Abstract

The implied volatility surface is a central object in hedging and risk management, each of which largely depends on how the surface will move rather than where it sits today. The literature leaves direct multi-step forecasting of the full surface from its own history under-addressed.

This work forecasts the S&P 500 surface multiple horizons ahead and asks whether respecting its structure improves forecasts. A smoothed surface dataset is built from two decades of option quotes, and a ladder of models is evaluated on it at matched capacity, from a linear baseline through plain transformers to the proposed Surface-Aware Neural Tensor Attention (SANTA) architecture. SANTA tokenises each surface cell, adds learned moneyness and maturity coordinate embeddings, and applies factored axial attention — separately across moneyness, across maturity, and across time — so that each cell attends along the surface’s two structural axes and its own history. The ladder is benchmarked against persistence and a polynomial-coefficient vector autoregression.

On point-wise error SANTA leads at the longer horizons, recording the lowest MSE and MAE, though by thin margins. The advantage is sharper on cross-sectional rank correlation — which scores the surface’s shape — where the SANTA variants separate clearly from the rest. Direct ablations locate this gain in the surface-aware coordinate embeddings, with spatial attention adding little thereafter.

Plain Language Summary

Financial markets offer contracts called options — a kind of insurance against future moves in prices like the stock market. Their cost reveals what investors expect about how turbulent the future will be. Taken together, these expectations form a snapshot that captures the market's view of risk at a single moment.

This snapshot shifts daily, and those who price contracts or guard against crashes care less about how it looks today than how it may look weeks or months ahead.

This project asks whether that snapshot can be forecast from its own recent history, and whether a forecasting tool built to understand its structure does better than one that ignores it.

Declaration of Originality

I hereby declare that the work presented in this thesis is my own unless otherwise stated. To the best of my knowledge the work is original and ideas developed in collaboration with others have been appropriately referenced. I acknowledge the use of a large language model (Claude Opus 4.8, by Anthropic) solely to improve the quality of English in this thesis.

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Acknowledgments

I would like to express my deepest gratitude to my supervisor Prof. Danilo Mandic and his PhD student Wuyang Zhou. Their guidance and patient feedback truly helped shape this work.

To my parents and my sister — thank you for your unwavering love and support. And to my friends, both here and back home, thank you for being a source of joy, and thank you for being there through it all.

Contents

Abstract	i
Plain Language Summary	ii
Declaration of Originality	iii
Copyright Declaration	iv
Acknowledgments	v
List of Acronyms	ix
List of Figures	x
1 Introduction	1
1.1 The Role of the Implied Volatility Surface	2
1.2 The Surface as a Structured Object	2
1.3 Research Question	3
1.4 Approach and Contributions	3
1.5 Thesis Structure	4
2 Background	5
2.1 Options and Implied Volatility	5
2.2 The Implied Volatility Surface	10
2.2.1 Static Structure	10
2.2.2 Dynamic Structure	12
2.3 Constructing the Surface from Market Quotes	14
2.3.1 The Construction Problem and Coordinate System	14
2.3.2 Construction Approaches and this Project's Choice	15
2.4 Forecasting the Implied Volatility Surface	15
2.4.1 Forecasting Through a Reduced Representation	16
2.4.2 Neural Approaches	16
2.4.3 Surface-Aware Transformers	17

3	Implementation	18
3.1	Data and Surface Construction	19
3.1.1	Data Source	19
3.1.2	Filtering and Quality Control	19
3.1.3	Coordinate System and Smoothing	20
3.1.4	The Resulting Surface	21
3.1.5	Diagnostics and Validation	23
3.2	The Forecasting Task	25
3.2.1	Train, Validation, Test Split	25
3.2.2	Target Space	26
3.2.3	Forecast Horizons	26
3.3	Classical Benchmarks	27
3.3.1	Persistence	27
3.3.2	The Polynomial-Coefficient VAR	28
3.4	Model Implementations	29
3.4.1	Day-Token Transformer	33
3.4.2	Per-Cell Transformer	35
3.4.3	Surface-Aware Embeddings	37
3.4.4	SANTA-Temporal	39
3.4.5	SANTA-Flat	40
3.4.6	SANTA	41
3.5	Evaluation Criteria	44
4	Results	46
4.1	Aggregate Forecasting Accuracy	47
4.2	Point-wise Error	47
4.2.1	Post-crisis Period	49
4.2.2	Crisis Period	51
4.3	Cross-sectional skill (IC)	53
4.4	The Cost of Unconstrained Spatial Attention	56
4.5	Persistence as an Anchor	57
4.6	Summary of Results	58

5	Evaluation	61
5.1	Objectives Revisited	61
5.2	Key Empirical Findings	62
5.3	Limitations	62
	Conclusions and future directions	64
A	Data Implementation Details	66
A.1	Data pipeline	66
A.2	Filter thresholds	67
A.3	Smoothing configuration	67
B	Model Architectural Details	69
C	Supplementary Results	71
C.1	Sample Forecasts	71
C.2	Crisis and Post-crisis Period Aggregate Performance Tables	76
C.3	Post-crisis MAE	78
C.4	Crisis-period MAE	79
C.5	Statistical Significance Supplementary Results	80
	Bibliography	84

List of Acronyms

ATM At-the-money

ITM In-the-money

OTM Out-of-the-money

SPX Standard & Poor's 500 Index

CBOE Chicago Board Options Exchange

VAR Vector Auto-regression

OLS Ordinary Least Squares

BIC Bayesian Information Criterion

SANTA Surface-Aware Neural Tensor Attention

VIX CBOE Volatility Index

IC Information Coefficient

MSE Mean Squared Error

MAE Mean Absolute Error

List of Figures

2.1	Payoff at expiry of a European call (left) and put (right) with strike K , as a function of the underlying price S_T	6
2.2	The Black–Scholes price of an option as a function of volatility σ , holding the other inputs fixed.	8
2.3	A representative implied volatility surface for an equity index, as a function of moneyness and maturity.	9
2.4	Implied volatility smile across strikes for an equity index	11
3.1	Number of SPX option records per trading day, 2 January 2004 to 29 December 2023.	19
3.2	Smoothed implied volatility surfaces on a sparse early-sample day and on a stressed COVID-period day.	22
3.3	Sample mean and standard deviation of the implied volatility surface across the grid.	24
3.4	At-the-money 30-day implied volatility across the full sample.	24
3.5	Train, validation, and test split shown against surface level and surface range.	25
3.6	End-to-end pipeline shared by every model.	30
3.7	Day-Token Transformer architecture.	35
3.8	Per-Cell Transformer architecture.	37
3.9	Surface-aware token assembly.	38
3.10	SANTA-Temporal architecture.	39
3.11	SANTA-Flat architecture.	41
3.12	SANTA architecture.	43
4.1	Concentration of forecast error over the test period.	49
4.2	Post-crisis forecast MSE.	50
4.3	Crisis-period forecast MSE.	52
4.4	Cross-sectional IC of the shape-reversion rule by forecast step.	54
4.5	Cross-sectional rank IC over the full test set.	55
4.6	Cross-seed forecast dispersion by horizon.	57
4.7	Per-step path error relative to persistence.	58
C.1	Sample forecast surface, $h + 10$	72
C.2	Sample forecast surface, $h + 21$	73

C.3	Sample forecast surface, $h + 42$	74
C.4	Sample forecast surface, $h + 63$	75
C.5	Post-crisis forecast MAE.	78
C.6	Crisis-period forecast MAE.	79

1

Introduction

Contents

1.1 The Role of the Implied Volatility Surface	2
1.2 The Surface as a Structured Object	2
1.3 Research Question	3
1.4 Approach and Contributions	3
1.5 Thesis Structure	4

The implied volatility surface is a central object in modern derivatives markets. At any given moment, it summarises how options across a range of strikes and maturities are priced relative to a single underlying, and its shape encodes market participants' collective expectations about future uncertainty. It is estimated continuously, in detail, and plays a direct role in the pricing and hedging of most derivative instruments.

The literature on volatility is correspondingly large, and its focus broad. Much of it is concerned with modelling the surface at a single moment — fitting it, smoothing it, ensuring it is free of arbitrage — or with explaining its shape through stochastic models of the underlying. Forecasting how the surface evolves is a smaller strand, and within it, direct statistical forecasting of the full surface across multiple horizons has received comparatively little attention. This project takes up that question directly, forecasting the full implied volatility surface across multiple horizons. Stated precisely, the question has two parts: *whether* models that respect the surface's structure outperform those that do not; *where* any advantage lies — at which horizons, under which market conditions, and on which aspect of the surface, its level or its shape.

1.1 The Role of the Implied Volatility Surface

Options are contracts whose value depends on the future behaviour of an underlying asset. Their prices therefore reflect what market participants currently believe about that behaviour, in particular how much the underlying might move. Implied volatility expresses each option's price as a single number on a common scale, and the implied volatility surface is the full cross-section of these numbers across strikes and maturities. It is at once a snapshot of how the option market is currently quoted and of how participants are pricing future movement in the underlying.

The surface is correspondingly central to a range of activities in derivatives markets. The most direct use is in pricing options that are not themselves quoted — illiquid strikes, bespoke maturities, or contracts traded over the counter — by interpolating their implied volatilities from nearby quoted points on the surface. Hedging goes further: a book of options carries exposure not to any single point on the surface but to how the surface as a whole shifts and twists, since it is these movements that change the value of the position. Risk management goes further still: stress testing and scenario analysis ask not only what the surface looks like today, but how it might deform or evolve tomorrow.

What links these last two is that they are not about today's surface at all, but about tomorrow's. Hedging over any horizon, and assessing how a portfolio may behave under future conditions, both require a view of how the surface itself will change — and that view is only as good as one's ability to forecast it.

1.2 The Surface as a Structured Object

The implied volatility surface, observed at a single moment, is a two-dimensional object: a grid indexed by the strike of the option and its time to maturity. That grid is structured. Neighbouring strikes carry similar information, and so do neighbouring maturities; the surface varies smoothly across both, so a point is closely related to those beside it and only loosely related to those far across the grid. Observed over a sequence of days, a third axis is added — time — along which consecutive surfaces are in turn highly correlated. The result is not just a high-dimensional time series but a doubly structured one: smooth across the grid at each moment, and persistent from one moment to the next.

This layout is invisible to a forecasting model that treats each surface as a flat vector. Clas-

sical and some modern multivariate time-series methods take as input a single sequence of high-dimensional vectors, and thus would employ no architectural distinction between two strikes that sit side by side on the surface and two that lie far apart. The spatial structure — which points are neighbours, and which are not — is discarded at the input stage; whatever the surface’s shape encodes is left for the model to recover from data alone.

An alternative is to keep the sequence of surfaces as a three-dimensional tensor, with separate axes for strike, maturity, and time. Methods designed for tensor-valued inputs can in principle exploit this organisation directly, treating variation across strikes, across maturities, and across time as related but distinct.

1.3 Research Question

The surface, then, is a structured object. The natural conjecture is that models respecting that structure should forecast it better. Whether the conjecture holds in practice is not obvious. Recent work in time-series forecasting has repeatedly found simple models, including plain linear ones, hard to beat, and added architectural complexity often fails to maintain its promise of improved performance out-of-sample. A structural assumption that is well-motivated in principle may be redundant when the surface’s recent history is already easy to extrapolate, and a liability when it is not.

There is no reason this should play out the same way in all market conditions. The surface does not evolve uniformly: in calm periods it moves smoothly, while under stress it can shift abruptly — and a model that wins in one regime need not win in the other. The question, then, is not simply whether structure helps on average, but where it helps and where it does not.

This project takes up that question directly, forecasting the full implied volatility surface across multiple horizons. The aim is to establish not only whether structure-aware models outperform, but over what horizons, and under which market conditions.

1.4 Approach and Contributions

This project serves as an empirical study: it implements a variety of forecasting models that differ in how much of the surface’s structure they encode, and benchmarks them under a single protocol.

The surface is not observed directly, so the target is first constructed by smoothing observed option-implied volatilities. The forecasting task asks how well the implemented models are able to forecast the full surface across all strikes and maturities from the surface’s own past.

Within this setting, the models form a ladder of increasing structural awareness. The deliverables for this project are therefore:

- the construction of a clean implied volatility surface dataset with a multi-horizon forecasting task defined on it;
- the design of a surface-aware architecture for forecasting the surface, evaluated against a controlled ladder of direct ablations;
- a systematic benchmark of this ladder against classical baselines, on the same data under a common multi-horizon protocol;
- an empirical analysis of whether and under what market conditions structural priors improve forecasting performance.

1.5 Thesis Structure

The remainder of this report is organised as follows. Chapter 2 reviews the background the project builds on: implied volatility and the surface it forms, the static and dynamic regularities the surface exhibits, the several surface construction methods, and existing approaches to forecasting it. Chapter 3 sets out the implementation, documenting the surface construction, defining the forecasting task, introducing the classical benchmarks, describing the ladder of models from the linear floor through to SANTA, and setting the evaluation metrics. Chapter 4 presents the empirical results, across forecast horizons and across calm and stressed regimes, isolating the contribution of each structural prior. Chapter 5 summarises the project’s findings, and evaluates it against its aims. The final chapter draws conclusions and proposes directions for future work.

2

Background

Contents

2.1 Options and Implied Volatility	5
2.2 The Implied Volatility Surface	10
2.2.1 Static Structure	10
2.2.2 Dynamic Structure	12
2.3 Constructing the Surface from Market Quotes	14
2.3.1 The Construction Problem and Coordinate System	14
2.3.2 Construction Approaches and this Project's Choice	15
2.4 Forecasting the Implied Volatility Surface	15
2.4.1 Forecasting Through a Reduced Representation	16
2.4.2 Neural Approaches	16
2.4.3 Surface-Aware Transformers	17

2.1 Options and Implied Volatility

An option is a contract that grants its holder the right, but not the obligation, to trade an underlying asset at a fixed price. This project is concerned with *European* options, which may be exercised only at a fixed expiry date.¹ A European *call* gives the right to buy the underlying at a strike price K at maturity T ; a European *put* gives the right to sell at K at T . Writing S_T for the price of the underlying at maturity, the holder exercises only when doing so is profitable, so the payoffs are

$$C_T = \max(S_T - K, 0), \quad P_T = \max(K - S_T, 0). \quad (2.1)$$

¹Many traded options are *American* and may be exercised at any time up to expiry, which complicates their valuation. The options studied in this work are European-style, so early exercise does not arise.

A call is exercised when the underlying finishes above the strike and is worthless otherwise; a put is exercised when it finishes below. An option whose strike sits on the exercise side of the underlying is said to be In-the-money (ITM); on the other side, Out-of-the-money (OTM); at the strike itself, At-the-money (ATM).

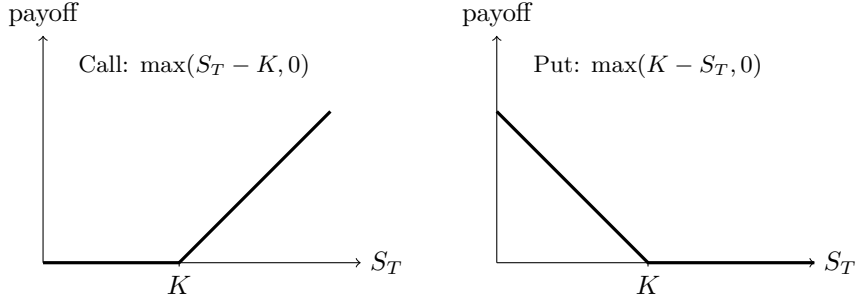


Figure 2.1: Payoff at expiry of a European call (left) and put (right) with strike K , as a function of the underlying price S_T .

The defining feature of these payoffs is their asymmetry. The loss to the holder is bounded below by zero, while the gain grows without limit as the underlying moves further into the money. A payoff of this shape is worth more when the underlying is more likely to make a large move, because large favourable moves are rewarded in full while large unfavourable moves are truncated at zero. The value of an option therefore depends not only on where the underlying is expected to end up, but on the entire range of outcomes it might reach — and, in particular, on how widely those outcomes are dispersed.

The value of an option today is the price a buyer should be willing to pay for that uncertain future payoff. Under the standard no-arbitrage theory of pricing, this value is the expected payoff, discounted back to the present, computed not under real-world probabilities but under the *risk-neutral* measure $\mathbb{Q}$ — the probabilities under which all traded assets earn the risk-free rate on average.² For a European call with strike K and maturity T , the value today is therefore

$$C_0 = e^{-rT} \mathbb{E}^{\mathbb{Q}}[\max(S_T - K, 0)], \quad (2.2)$$

where r is the risk-free rate and the expectation is taken over the distribution of the terminal price S_T .

Equation (2.2) makes precise why dispersion matters. Because the payoff is a convex function of S_T , spreading the distribution of S_T while holding its mean fixed can only increase the expected

²The existence of such a measure is equivalent to the absence of arbitrage; its use is what makes discounted prices consistent across instruments. We take it as given and do not develop the theory here.

payoff.³ An option is thus, in large part, a position on how widely the underlying might move. To turn Equation (2.2) into a concrete price, however, one needs a model for the distribution of S_T — and it is here that the Black-Scholes framework enters.

The Black-Scholes model [1] specifies the dynamics of the underlying. It assumes that the underlying price follows a geometric Brownian motion with constant volatility,

$$dS_t = r S_t dt + \sigma S_t dW_t, \quad (2.3)$$

where W_t is a standard Brownian motion and the single parameter σ governs the magnitude of the underlying's random fluctuations. Under this assumption the terminal price S_T is log-normally distributed, and the expectation in Equation (2.2) can be evaluated in closed form. The resulting price of a European call is

$$C_{\text{BS}}(S, K, T, r, \sigma) = S N(d_1) - K e^{-rT} N(d_2), \quad (2.4)$$

where $N(\cdot)$ is the standard normal cumulative distribution function and

$$d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}. \quad (2.5)$$

Of the five quantities on which the Black-Scholes price depends, four — the underlying price S , the strike K , the maturity T , and the risk-free rate r — are directly observable. Only the volatility σ is not. The formula should therefore be read, for the purposes of this work, as a function that maps a volatility to a price: given σ , it returns the model price of an option. Its assumptions — a single constant volatility, continuous price paths with no jumps, and frictionless trading — are idealisations, and the consequences of their failure are what the remainder of this section builds towards.

This map can be inverted. An option trading in the market has an observed price; the question is what value of σ , substituted into the Black-Scholes formula, reproduces that price exactly. This value is the option's *implied volatility*, σ_{IV} , defined implicitly by

$$C_{\text{BS}}(S, K, T, r, \sigma_{\text{IV}}) = C_{\text{market}}, \quad (2.6)$$

where C_{market} is the observed market price of the option.

³Again, the truncation at zero caps the effect of unfavourable outcomes, while favourable ones contribute in full.

That this equation has a unique solution is not automatic; it follows from a property of the Black-Scholes price. Differentiating Equation (2.4) with respect to volatility gives

$$\frac{\partial C_{\text{BS}}}{\partial \sigma} = S \phi(d_1) \sqrt{T} > 0, \quad (2.7)$$

where $\phi(\cdot)$ is the standard normal density. The price is therefore strictly increasing in σ : a higher volatility always implies a higher option price. Equation (2.6) thus has at most one solution, and for any market price consistent with no-arbitrage bounds it has exactly one. Figure 2.2 illustrates this: the Black-Scholes price traces a strictly increasing curve in σ , and the observed market price selects a single point on it.

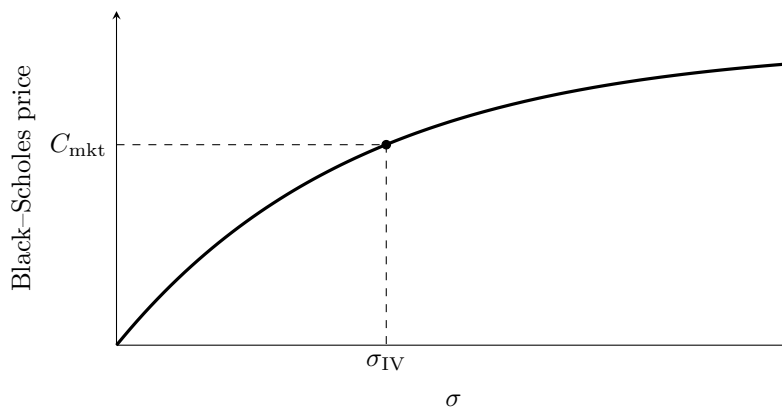


Figure 2.2: The Black-Scholes price of an option as a function of volatility σ , holding the other inputs fixed.

To clarify, implied volatility is not a direct forecast of the underlying's future volatility, nor a quantity measured directly from the market; it is the number that, fed through a model known to be imperfect, recovers the option's price. In effect, implied volatility is a change of units. It restates a price — which depends on strike, maturity, and the level of the underlying — as a single number on a common scale, on which the prices of different options can be set side by side.

A single option yields a single implied volatility. Applying the inversion of Equation (2.6) to every option on the same underlying yields a collection of them, one for each combination of strike and maturity traded. If the Black-Scholes model were an accurate description of the market, this collection would be trivial: a single constant volatility would price every option, and the same σ_{IV} would be recovered from each. This is not what is observed. Implied volatility varies systematically with both the strike and the maturity of the option from which it is computed.

Collecting these values gives a function of the two contract characteristics,

$$\sigma_{\text{IV}} : (K, T) \mapsto \sigma_{\text{IV}}(K, T), \quad (2.8)$$

known as the *implied volatility surface*. At a fixed maturity, its variation across strikes is the volatility *smile* or *skew*; at a fixed strike, its variation across maturities is the *term structure* of implied volatility. For an equity index such as the one studied here, the surface is characteristically downward-sloping in strike and flattens as maturity lengthens. A representative surface is shown in Figure 2.3.

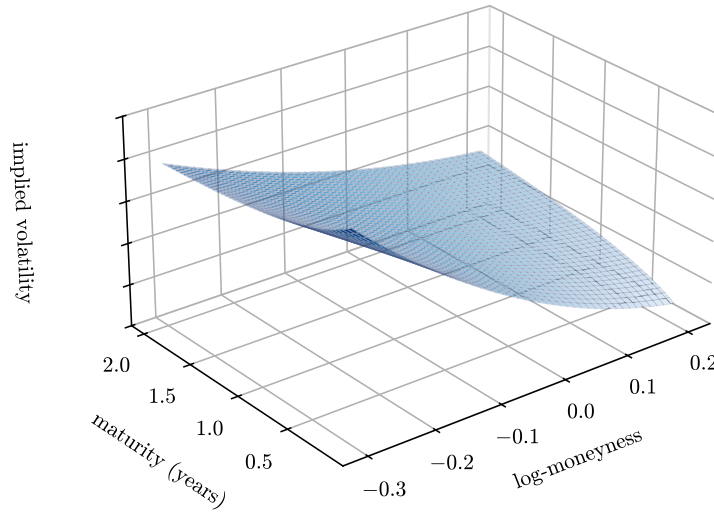


Figure 2.3: A representative implied volatility surface for an equity index, as a function of moneyness and maturity.

In practice the surface is rarely expressed in terms of the raw strike. To make surfaces comparable across days and underlyings, the strike axis is re-expressed in terms of *moneyness* — the strike relative to the current price of the underlying — and, in this work, relative to the forward price rather than the spot.

2.2 The Implied Volatility Surface

2.2.1 Static Structure

Section 2.1 established that the implied volatility surface is not flat. Its departures from flatness are not arbitrary; they follow regular, well-documented patterns along both of its axes.

Variation across strike: the smile and the skew

Holding maturity fixed and varying the strike, implied volatility traces a curve known as the smile — so called because in some markets it turns up at both ends. For equity indices the curve is not symmetric but downward-sloping: implied volatility is systematically higher for low-strike options than for high-strike options. This asymmetry is called the volatility skew, and it is a persistent, pronounced feature of index markets rather than an occasional one (Figure 2.4).

The skew forms due to the underlying's actual behaviour. Index returns are not symmetric: large downward moves are bigger and more sudden than large upward ones, and crashes tend to coincide with spikes in volatility — when equity markets fall, they fall faster than they rise. Low-strike options are precisely the ones that pay off when the index drops sharply, so the market, knowing those outcomes are more likely and more violent, charges more for them — and a higher price, expressed on the implied-volatility scale, is a higher implied volatility.

In this sense, low-strike options act as insurance against a falling market. Naturally, the demand is one-directional — almost everyone wants downside protection, few want to sell it — and that imbalance bids up the price of low-strike options, and with it their implied volatility.

Both effects have a single unifying interpretation: the surface implies a probability distribution for where the underlying will end up,⁴ and the skew is the surface's way of saying that distribution has a fatter left tail — more weight on bad outcomes — than Black–Scholes' log-normally distributed assumption would give it.

Variation across maturity: the term structure

Fixing the level of moneyness and varying the time to expiry instead, implied volatility changes again, in a manner described by the term structure of volatility. Its shape is itself time-varying: in

⁴This distribution is made precise below, in the discussion of butterfly arbitrage.

calm periods it usually slopes upward, near-term options implying less volatility than longer-dated ones, while in periods of acute short-term stress it can invert, the market pricing high turbulence now that it expects to subside.

Figure 2.4 shows a typical equity index surface structure: several smile slices at increasing maturities for an equity index, each downward-sloping and flattening as maturity lengthens.

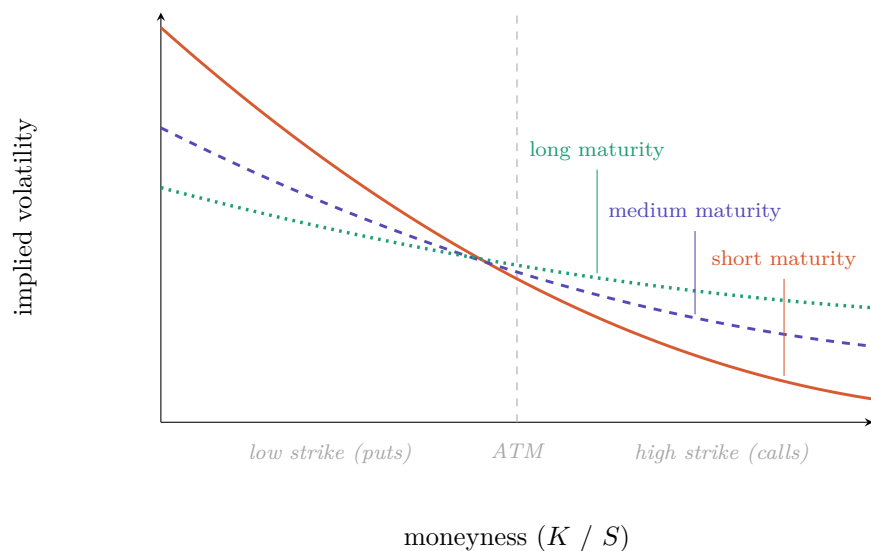


Figure 2.4: The implied volatility smile for an equity index, shown at three maturities. Across each fixed maturity, implied volatility declines from low strikes to high strikes. The skew is most pronounced at short maturities and flattens as maturity lengthens.

Absence of arbitrage: the shape is constrained

The surface cannot take just any shape. Because its implied volatilities are derived from the prices of related options on the same underlying, those prices must be mutually consistent in a specific sense: it must be impossible to combine the options into a portfolio that costs nothing to set up yet cannot lose money. Such a portfolio would be an arbitrage, and in a functioning market it cannot persist — traders would exploit it without limit until prices moved. A surface that permitted one would not describe any real market. Two consistency conditions are standard.

The first, absence of calendar-spread arbitrage, constrains how implied volatility may vary across maturities at a fixed strike: total implied variance — implied volatility squared, scaled by time to expiry ($\sigma_{IV}^2 T$) — must not decrease as expiry lengthens. The intuition is that a longer-dated option offers everything a shorter-dated one of the same strike does and more, because it has more time; were its total variance lower, the two could be traded against each other for a riskless profit.

The second, absence of butterfly arbitrage, constrains how implied volatility may vary across strikes at a fixed maturity. It has an elegant equivalent: the same option prices that define the surface also pin down the probability distribution the market assigns to the underlying at expiry: that risk-neutral density is recovered from the call price C as its second derivative in the strike K ,

$$q(K) = e^{rT} \frac{\partial^2 C}{\partial K^2}, \quad (2.9)$$

a classical result due to Breeden and Litzenberger [2]. No-butterfly-arbitrage is simply the requirement that this density be a genuine one — that $q(K) \geq 0$ for every strike. A surface that violates the condition implies $q(K) < 0$ somewhere, claiming some outcome occurs less often than never, and that negativity is exactly what a butterfly position⁵ would monetise into a riskless profit.

This project is concerned with forecasting the surface as the market quotes it, without imposing no-arbitrage conditions on either the constructed surfaces or the forecasts. The reasoning behind setting them aside, and how that choice relates to the construction method used here are taken up in Section 2.3.2.

2.2.2 Dynamic Structure

The smoothness of the surface across strike and maturity shapes how it moves: neighbouring points carry similar information, so the surface does not evolve as a collection of independent series but as a coordinated whole, its daily changes concentrated in a small number of common modes.

Early principal-component studies applied the technique to the smile at individual maturities, finding two to three components sufficient to describe most of its variation [3]. Extending the decomposition across the surface, Fengler, Härdle and Villa [4] showed via common principal components that a small number of factors — a shift, a Z-shaped slope, and a twist — are shared across maturity slices. Cont and da Fonseca [5] decomposed the daily changes of log-implied-volatility for the surface as a whole, identifying an overall shift in level, a tilt of the skew, and a change in curvature, with the level shift accounting for around 94% of the daily variance.

This concerns the variance of daily changes, not their predictability. Most of the surface's movement is concentrated in a few directions, but that a few components capture most of the variation does not mean the rest is unforecastable noise, nor that forecasting those components suffices to forecast the surface.

⁵Buying one option each at a low and a high strike while selling two at the strike midway between them; the combination costs little and cannot pay out less than zero, so a negative implied density would price it below nothing.

The surface’s dynamics hold two tendencies in tension. Over short horizons implied volatility is highly persistent, so tomorrow’s surface closely resembles today’s and the current surface is a powerful near-term predictor; over longer horizons it mean-reverts toward a long-run level. This reversion is not the level’s alone: the skew and curvature of the surface revert too [5]. The speed of that level reversion is not agreed upon: it has been estimated at around a month for the level factor and described as neither ‘slow’ nor ‘fast’ [5], as slow and near-unit-root for at-the-money implied-volatility series [6], and with an autoregressive coefficient around 0.9 for the fitted level coefficient [7]. The disagreement is itself the point, because the balance between persistence and reversion is not a fixed constant but shifts with the market environment — and so does the structure of the dynamics. The surface’s response to a given index move differs sharply between calm and stressed conditions [8], the predictive relationship between implied and realised volatility is statistically unstable over time [6], and forecasting methods that perform well in one period can degrade sharply across structural breaks, with factor-based models in particular losing stability out of sample while tree-based forecasts prove more robust [9]. A characterisation calibrated to one regime need not transfer to another, which is why evaluation across regimes, rather than within a single benign period, is informative.

That the dynamics are nonetheless partially predictable is well established. The surface can be forecast with statistical significance — for index and for individual-equity options alike — whether by modelling the joint dynamics of its cross-sectional coefficients [7], [10] or by applying machine-learning methods directly to the surface [9]. Two qualifications recur. Statistical predictability is robust but its economic value is fragile: the trading profits it implies survive only under low transaction costs, largely vanishing once realistic frictions are imposed [7], [10]. And the predictive gains come from exploiting structure — jointly modelling the whole surface outperforms forecasting any single point in isolation, and drawing on the market-index surface improves forecasts of individual-equity surfaces further still [10].

One regularity is deliberately set aside in this work. Movements in the surface are strongly and negatively correlated with returns on the underlying: implied volatility tends to rise when the index falls, an asymmetry often termed the *leverage effect*, visible in the decompositions above as a strong negative correlation between the level factor and the index return while the higher modes remain largely uncorrelated with it [3], [4], [5]. This is a relationship between the surface and an external variable, and exploiting it requires conditioning on that variable, as models incorporating the index return and the CBOE Volatility Index (VIX)⁶ do [8]. Because this project forecasts

⁶The VIX is a single index of the S&P 500’s expected thirty-day volatility, derived from a strip of index option prices.

the surface unconditionally, from its own history alone, the leverage effect lies outside its scope by construction, and the resulting forecasts can be read as a lower bound on what a conditional model could achieve. Taken together, these features describe a surface whose movements are structured but not reducible to any single, fixed, low-dimensional account holding across all conditions — and whose forecastability, given the role of the regime, may itself differ from one period to the next.

2.3 Constructing the Surface from Market Quotes

2.3.1 The Construction Problem and Coordinate System

The implied volatility surface is a continuous object, but the market does not quote it as one. On any given day, only a finite set of options are traded, each at a particular strike and maturity, and inverting the Black–Scholes formula at each yields implied volatilities at a scattered, irregular collection of points. The set of traded strikes and maturities also changes from day to day as contracts expire and new ones are listed. Constructing the surface is the problem of passing from this discrete, irregular, time-varying set of points to a continuous function defined on a common domain.

Before construction, it is convenient to fix the coordinate system. Section 2.1 used the spot price for clarity, but for an index paying dividends the natural reference is the forward price. For a maturity τ , the forward is

$$F_t(\tau) = S_t e^{(r-q)\tau}, \quad (2.10)$$

where r is the risk-free rate and q the dividend yield. Quoting options relative to the forward rather than the spot removes the deterministic effect of carry, so that surfaces on different days — with different spot levels, rates, and dividends — become directly comparable. In place of the raw strike, the surface is expressed in terms of log-forward-moneyness,

$$m = \ln\left(\frac{K}{F_t(\tau)}\right), \quad (2.11)$$

which is zero for an at-the-money-forward option, negative for low strikes, and positive for high strikes. For a call, positive m is OTM and negative m is ITM; for a put, the reverse. Expressing the surface as a function $\sigma(m, \tau)$ of log-forward-moneyness and maturity centres it on a common, dividend-adjusted scale.⁷ This is the convention adopted throughout this work.

⁷The same forward relationship underlies put-call parity [11], which links the prices of puts and calls at a common

2.3.2 Construction Approaches and this Project's Choice

Construction methods divide into two families: parametric methods, which impose an explicit functional form for the surface and fit its parameters to the observed prices, and non-parametric methods, which impose no such form and estimate the surface directly from the data.

The most widely used parametric method is the stochastic-volatility-inspired (SVI) parameterisation of Gatheral [12] for a single maturity, extended to a full surface by Gatheral and Jacquier [13] in a form that can be constrained to be free of static arbitrage; the quadratic specification of Dumas, Fleming and Whaley [14], linear in maturity and quadratic in moneyness, is a long-standing simpler alternative that carries no such arbitrage-free guarantee.

This project uses a non-parametric method that does not enforce the absence of arbitrage: the Nadaraya–Watson estimator [15], [16], which estimates implied volatility at a point (m, τ) as a Gaussian-kernel-weighted average of the observed implied volatilities,

$$\hat{\sigma}(m, \tau) = \frac{\sum_i \sigma(m_i, \tau_i) g(m - m_i, \tau - \tau_i)}{\sum_i g(m - m_i, \tau - \tau_i)}, \quad g(x, y) = \exp\left(-\frac{x^2}{2h_1}\right) \exp\left(-\frac{y^2}{2h_2}\right), \quad (2.12)$$

with bandwidths h_1, h_2 controlling the degree of smoothing across moneyness and maturity respectively. Two considerations motivate this choice. The first is the absence of a constrained form. Because the estimator imposes none, the representation does not confound the shape of the surface with the fit of a chosen parametric model, nor does it correct the surface toward imposing the no-arbitrage conditions, and this is deliberate. The object to be forecast is the market-made surface, including whatever small inconsistencies it contains. Projecting it onto a functional or arbitrage-free set would alter the very dynamics under study. The second is precedent: kernel-based representations of the market-made surface underlie the empirical study of surface dynamics in Cont and da Fonseca [5] and the recent generative model of Vuletić and Cont [17], among many others, placing the present work alongside an established literature.

2.4 Forecasting the Implied Volatility Surface

Forecasting volatility is a mature field, but the overwhelming majority of it targets a single number rather than a surface. Its classical core models the conditional variance of the underlying's returns directly — a real-world, scalar quantity, and a distinct object from the option-implied surface strike and maturity and allows either to be used in constructing the surface.

forecast in this work — beginning with the ARCH and GARCH families [18], [19] and the long line of descendants that followed.⁸ The other targets that dominate the field are scalar in the same way: realised volatility, an ex-post measure built from high-frequency returns, and the VIX. The implied volatility surface is higher-dimensional — a full cross-section across strike and maturity, evolving in time — and forecasting it is a correspondingly different problem.

2.4.1 Forecasting Through a Reduced Representation

The longest-established approach forecasts not the surface itself but a handful of numbers that summarise it: a representation (typically principal components or the coefficients of a parametric form) is fitted to each day’s surface, its low-dimensional time series is modelled, and the surface is reconstructed from the result.

The early factor decompositions of Section 2.2.2 were used to characterise surface dynamics: Cont and da Fonseca [5] fit autoregressive processes to their principal-component coefficients, and Fengler, Härdle and Mammen [20] a vector autoregression to their factor loadings — in each case to describe how the reduced series move in sample, not to forecast them out of it.

The parametric-coefficient models put the same design to work out of sample, replacing the factor basis with explicit functional forms. Gonçalves and Guidolin [7] fit a polynomial in moneyness and maturity to each daily cross-section and forecast its coefficients with a vector autoregression — an approach Bernales and Guidolin [10] extend across individual-equity surfaces — while Audrino and Colangelo [9] forecast a parameterised surface with boosted regression trees. Their out-of-sample skill is reported to be statistically significant.

Of these forecasters, the polynomial-coefficient vector autoregression of Gonçalves and Guidolin is the most widely studied instance of the design and it serves here as a classical benchmark.

2.4.2 Neural Approaches

A smaller and more recent body of work forecasts the full surface with neural networks, and these are the closest antecedents of the present work. Kim et al. [21] forecast one step ahead with a physics-informed convolutional architecture that embeds the Black–Scholes relation in its training objective. Zhang et al. [22] forecast a sampled surface together with a set of learned features,

⁸The conditional variance is the variance of the next return given current information; the GARCH family models it as a function of past returns and past conditional variances.

their main contribution being a deep network that reconstructs an arbitrage-free surface from the prediction.

A related strand simulates the surface rather than forecasting it. Generative models such as the variational-autoencoder approach of Ning et al. [23] and the generative-adversarial VolGAN of Vuletić and Cont [17] produce distributions of plausible future surfaces for risk-management and scenario use, rather than a point forecast of the next surface measured against its realised value.

2.4.3 Surface-Aware Transformers

The properties established in Section 2.2.2 point to what a forecaster of the surface should be able to do. The surface moves as a coordinated whole, its changes spanning strikes and maturities together, so a model able to let any region inform any other is a natural fit. Its dynamics are regime-dependent, so a model whose combination of information can adapt to the state of the surface, rather than stay fixed, is better suited than one that cannot. And it is persistent with long memory, so weighing a long history by relevance rather than by a fixed decay should be an advantage. Attention is a natural candidate: it learns, from the input itself, which elements of a structured object should inform which others, and it has reshaped sequence and multivariate forecasting in recent years [24], [25], [26], [27]. How much of this promise translates to the implied volatility surface is an empirical question.

Its use for forecasting the implied volatility surface remains thin. What work exists is largely confined to one-step-ahead prediction; the literature reviewed for this project surfaced no attention-based model forecasting the full surface multi-step from its own history.

This project takes up that setting. Rather than proposing a single architecture, it places a family of them — from plain transformers to ones that attend across the surface’s strike and maturity structure — in a controlled comparison, so that whatever each structural element contributes can be measured rather than assumed.

3

Implementation

Contents

3.1 Data and Surface Construction	19
3.1.1 Data Source	19
3.1.2 Filtering and Quality Control	19
3.1.3 Coordinate System and Smoothing	20
3.1.4 The Resulting Surface	21
3.1.5 Diagnostics and Validation	23
3.2 The Forecasting Task	25
3.2.1 Train, Validation, Test Split	25
3.2.2 Target Space	26
3.2.3 Forecast Horizons	26
3.3 Classical Benchmarks	27
3.3.1 Persistence	27
3.3.2 The Polynomial-Coefficient VAR	28
3.4 Model Implementations	29
3.4.1 Day-Token Transformer	33
3.4.2 Per-Cell Transformer	35
3.4.3 Surface-Aware Embeddings	37
3.4.4 SANTA-Temporal	39
3.4.5 SANTA-Flat	40
3.4.6 SANTA	41
3.5 Evaluation Criteria	44

3.1 Data and Surface Construction

3.1.1 Data Source

The data for this work are Standard & Poor’s 500 Index (SPX) option quotes from the OptionMetrics IvyDB US database, accessed via the Wharton Research Data Services (WRDS) platform. SPX options are European-style and cash-settled on the index,¹ traded on the Chicago Board Options Exchange (CBOE). The sample spans all 5,033 trading days from 2 January 2004 through 29 December 2023, and contains approximately 37 million option-record observations. The daily record count grows steadily over the sample, from a few hundred contracts a day in 2004 to around twenty thousand by 2023, reflecting the steady expansion of SPX option-quoting activity over the period; the full record is shown in Figure 3.1.

The implied volatilities used throughout this work are those supplied by OptionMetrics, computed from end-of-day mid-quotes of each contract by inverting the Black formula on the implied forward.² The data pipeline — field selection, file joining, and edge-case handling — is documented in Appendix A.1.

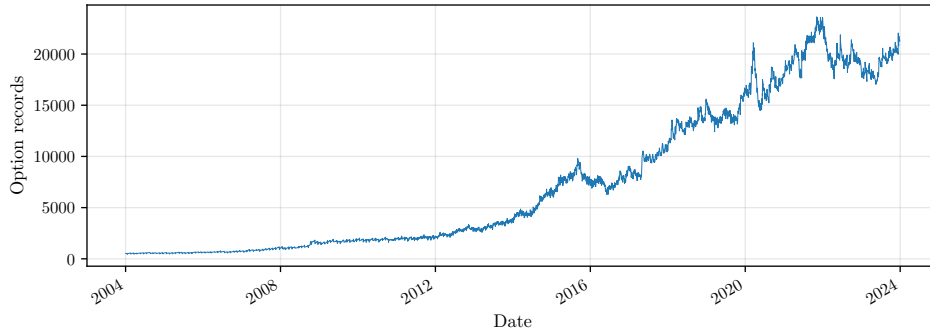


Figure 3.1: Number of SPX option records per trading day, 2 January 2004 to 29 December 2023.

3.1.2 Filtering and Quality Control

Four families of filter reduce the raw records to the set used for surface construction: scope, validity, liquidity, and the OTM convention. Exact thresholds are tabulated in Appendix A.2; the role of each is described here.

¹At expiry the option pays a cash amount equal to the difference between the index level and the strike, rather than delivering the underlying. Cash settlement is the standard since the index itself is not a tradeable security; equity options on individual stocks, by contrast, typically settle by physical delivery of the shares.

²The Black [28] formula is the standard Black–Scholes pricing equation [1] (introduced in Section 2.1) rewritten in terms of the forward price of the underlying rather than its spot. For a European option, the two are equivalent.

Scope filters define the range of contracts considered at all. The moneyness and maturity windows are set generous enough to cover the eventual target grid (Section 3.1.4) with substantial margin on every side, so the smoother has neighbouring quotes near the grid’s edges.

Validity filters remove records that are not interpretable as quotes: implied volatility and vega must be finite, and the implied volatility must lie within a wide bracket that excludes the sentinel values OptionMetrics records when its inversion fails.

Liquidity filters retain quotes that carry information. A vega floor excludes contracts so deep OTM or so close to expiry that small price errors invert to large volatility errors, and a positive-volume requirement excludes stale unfilled quotes. Applied uniformly, both requirements would leave the wings of the surface — the deep-OTM strikes on either side of the forward, where trading interest thins fastest — so sparsely supported that the smoother could not estimate them. The vega floor is therefore lowered in the wings and the volume requirement waived. The dense core can afford to be strict; in the wings, an imperfect quote is more useful than no quote at all.

The OTM convention takes the cleaner side of each strike: calls are kept for $m \geq 0$, puts for $m \leq 0$, with a narrow band around the forward admitting both types so the ATM-forward column is served by either. OTM quotes carry only time value and are more precisely measured than their ITM counterparts; put-call parity makes the choice information-preserving in theory, and the gain is measurement quality in practice.

3.1.3 Coordinate System and Smoothing

The Nadaraya–Watson estimator of Section 2.3.2 is applied in the vega-weighted form of [5], [17]: each quote contributes in proportion to its vega, the partial derivative of the option’s price with respect to its implied volatility,³ so that quotes whose prices are more sensitive to the volatility signal carry more weight:

$$\hat{\sigma}(m, \tau) = \frac{\sum_i \nu_i \sigma_i g(x_i, y_i)}{\sum_i \nu_i g(x_i, y_i)}, \quad x_i = \log \tau - \log \tau_i, \quad y_i = m - m_i, \quad (3.1)$$

where g is the Gaussian kernel of equation (2.12), here with bandwidth $h_1 = h_\tau$ acting on the log-maturity difference x_i and $h_2 = h_m$ on the moneyness difference y_i , and ν_i the vega of quote i . The implementation departs from the standard form in two places: the kernel acts in $(\log \tau, m)$ rather than (τ, m) , and the moneyness bandwidth varies with maturity.

³Contracts with high vega have prices that respond strongly to the volatility signal and so yield more precisely identified implied volatilities when the inversion is performed.

The $\log \tau$ coordinate reflects the multiplicative scale on which maturities relate. A Gaussian kernel measures distances symmetrically in its inputs, so kernel distance in $\log \tau$ corresponds to ratio in τ : a thirty- to sixty-day pair and a six- to twelve-month pair are treated as equally close, both being a ratio of two.

The maturity bandwidth h_τ is held constant across the surface; the moneyness bandwidth h_m is not. At the short end, where quotes are dense and the smile is sharply resolved, a tight h_m recovers curvature cleanly. At the long end, where quotes thin out and the smile is shallower, the same tight h_m would leave too few quotes contributing to each target moneyness. The moneyness bandwidth is therefore ramped wider as $\log \tau$ grows, with the specific values and the ramp profile in Appendix A.3.

Some cells may still be served by too few quotes, and two diagnostics guard against this. The first is the effective sample size,

$$n_{\text{eff}} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2}, \quad (3.2)$$

where $w_i = \nu_i g(x_i, y_i)$; this is required to exceed 2, ensuring no cell is estimated from a single quote. The second is the maximum single-quote weight share, $\max_i w_i / \sum_i w_i$, required to stay below 0.85. Cells failing either guard are marked *unsupported* and filled by their immediate grid neighbours.

3.1.4 The Resulting Surface

The smoothed surface is evaluated on a fixed grid of eleven moneyness points and ten maturity points, producing one hundred and ten cells per trading day.

The moneyness axis covers $m \in [-0.10, +0.10]$ in uniform steps of 0.02:

$$m \in \{-0.10, -0.08, -0.06, -0.04, -0.02, 0, +0.02, +0.04, +0.06, +0.08, +0.10\}.$$

The bounds span the bulk of liquid OTM quotes for SPX index options over the sample, while remaining narrow enough that the smoother is supported throughout. The odd point count places the centre cell exactly at the at-the-money-forward, $m = 0$.

The maturity axis covers ten maturities in calendar days:

$$\tau \in \{30, 45, 60, 90, 120, 150, 180, 240, 300, 365\} \text{ days.}$$

The endpoints define the range over which the surface is most actively traded: for maturities shorter than thirty days, settlement effects dominate, and for maturities longer than a year, quote density drops sharply. The spacing is non-uniform — finer at the short end, coarser at the long end — matching where the surface moves fastest.

The resulting daily surface is a matrix

$$S_t \in \mathbb{R}^{11 \times 10}, \quad S_t(j, k) = \hat{\sigma}(m_j, \tau_k; t), \quad (3.3)$$

indexed over the eleven moneyness points m_j and ten maturities τ_k , with one such matrix produced for each of the 5,033 trading days in the sample.

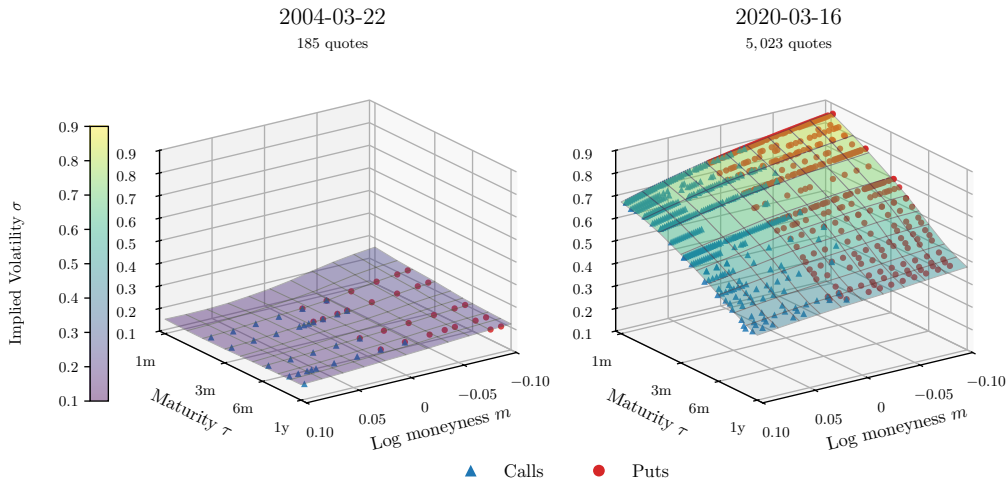


Figure 3.2: Smoothed implied volatility surfaces on a sparse early-sample day (*left*, 22 March 2004) and on a stressed COVID-period day (*right*, 16 March 2020), with the day’s retained call and put quotes overlaid as scatter points. Quote counts above each panel are post-filter; the points plotted are those falling within the displayed grid window.

Figure 3.2 shows the construction on two trading days at opposite ends of the sample. 22 March 2004, an early-sample day, has 185 retained quotes; the surface sits low and nearly flat, with shallow term structure and shallow negative skew. 16 March 2020, a stressed COVID-period day on which the S&P 500 fell by 12.0%,⁴ has 5,023 retained quotes; the surface carries a sharp negative skew, pronounced term structure, and a much higher overall level. In both cases the retained quotes lie on the smoothed surface, and SPX’s characteristic negative skew is preserved, illustrating that the construction holds up across both sparse and dense quote support and across calm and stressed conditions.

⁴The largest single-day decline in the index since 1987.

3.1.5 Diagnostics and Validation

Statistical fit

The median per-day root-mean-squared error between the gridded surface and the day's retained quotes is 0.0044 in implied volatility, against typical SPX implied volatilities in the 15%-25% range, and is 0.0037 at the ATM core versus 0.0047 in the wings.⁵ The wing figure is larger, as expected — quote density is lowest there and the smoother is doing the most work to bridge the gaps.

Grid support

Support across the grid is near-uniform. Of the 553,630 cell-days in the sample (5,033 trading days $\times$ 110 cells), only eight require neighbour-fill, all in the same cell at $\tau = 30\text{d}$, $m = +0.10$ — the corner of the surface where quote density is lowest.

Arbitrage violations

As discussed in Section 2.3.2, the construction does not enforce static no-arbitrage. Whether the resulting surfaces are nonetheless valid is checked by computing the two standard conditions — calendar-spread across maturities and butterfly across strikes — on each constructed surface.

Both kinds of violation are rare and small. Butterfly violations, checked across strikes at each maturity, appear on 0.52% of maturity-days; calendar-spread violations, checked across maturities at each moneyness, are rarer, on 0.018% of moneyness-days. Both sit within typical bid-ask spreads in magnitude, though calendar violations, where they occur, are the larger of the two.

Sample-wide behaviour

Taken across the full sample, the surface ensemble has the shape a recognisable SPX implied-volatility surface should. The mean surface (Figure 3.3, left) carries a pronounced negative skew at the short end — the downside wing sits roughly 8% of implied volatility above the at-the-money level at 30 days, against under 2% on the upside — and the skew flattens out by the long end, by which point the two wings sit nearly equidistant from the ATM level. The ATM level itself rises

⁵ATM and wings are defined here as $|m| < 0.03$ and $|m| \geq 0.03$ respectively.

from 16.6% at 30 days to 18.9% at one year, giving the mild upward term structure characteristic of SPX.

The cell-level standard deviation across the sample (right panel) is concentrated in the short-dated near-ATM band, peaking at 8.0% at the 30-day ATM cell. This is the part of the surface that moves most: the at-the-money short-dated level responds fastest to market events, and the wings, while large in absolute terms, vary slightly less day-to-day. By a year out, the day-to-day variability across the entire moneyness range is roughly two-thirds that of the short end.

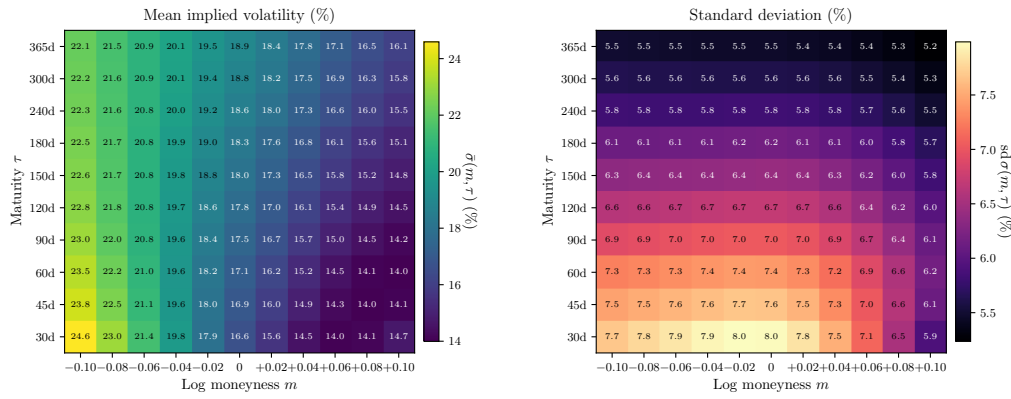


Figure 3.3: Per-cell mean implied volatility (*left*) and standard deviation (*right*) across the 5,033 trading days of the sample.

The time evolution of the surface tracks the major volatility regimes of the period (Figure 3.4). The global financial crisis pushes the ATM 30-day series to its first sharp peak in late 2008; the COVID drawdown of March 2020 produces a comparable but sharper spike; and the 2022 inflation regime sustains elevated levels through that year. Taken together with the fit, support, and arbitrage checks above, the surfaces are sound for the work that follows.

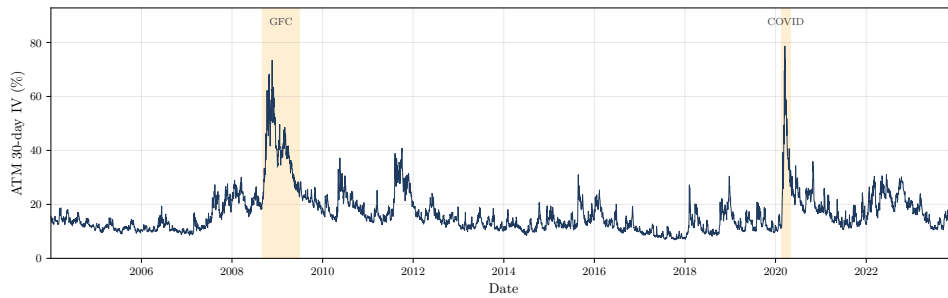


Figure 3.4: At-the-money 30-day implied volatility, 2 January 2004 to 29 December 2023.

3.2 The Forecasting Task

The constructed surfaces of Section 3.1 form a time series of daily 11×10 tensors. The task built on them is multi-step forecasting: predicting a path of future surfaces from past surfaces.

3.2.1 Train, Validation, Test Split

The full sample of 5,033 trading days, from 2 January 2004 to 29 December 2023, is split chronologically in proportions 70/10/20: 3,523 training days (to the end of 2017), 503 validation days (across 2018 and 2019), and 1,007 test days (2020 onwards).⁶

Figure 3.5 shows where each split falls against two daily summaries of the surface. The upper panel is the at-the-money 30-day implied volatility, the most natural single index of the surface’s *level*. The lower panel is the range of implied volatility across cells on each day — the highest minus the lowest cell — a direct measure of how stretched the surface is.

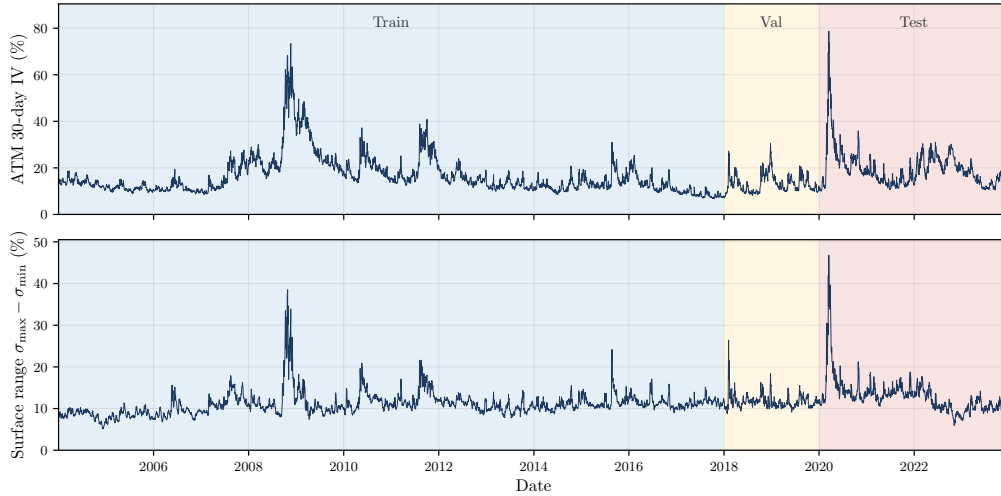


Figure 3.5: Train, validation, and test periods shaded against (*top*) ATM 30-day implied volatility, and (*bottom*) the range of implied volatility across cells on each day.

The split is deliberately constructed so that each block contains at least one period of pronounced market stress. The global financial crisis sits inside training, a large spike in both panels of the figure across late 2008. The early-2018 volatility spike sits inside validation; in level it is modest, but it shows clearly in the lower panel, a reminder that the surface can stretch sharply without its level rising much. The COVID drawdown of March 2020 sits inside test, the largest

⁶The split is by trading-day count rather than year-end, so the train-validation and validation-test boundaries fall on 31 December 2017 and 31 December 2019 respectively.

spike in both panels across the full sample, joined later in the test period by the elevated regime of 2022.

3.2.2 Target Space

Models forecast log implied volatility. The dynamics literature reviewed in Section 2.2.2 works in log-space throughout: shocks to implied volatility are multiplicative rather than additive, so a one-percentage-point IV move means something quite different at a 10% level than at an 80% level, and log-space absorbs the difference. A side benefit is that forecasts are positive by construction.

Cells are then standardised. Each cell’s log-IV series in the training set is rescaled to zero mean and unit variance, and the same training-set statistics are applied unchanged to the validation and test sets. The standardisation matters because the natural variability of the surface differs by cell. Section 3.1.5 reported standard deviations ranging from 5.2% at the long-dated upside wing to 8.0% at the short-dated near-ATM cell. Without per-cell scaling, any uniform loss would be dominated by the most-variable cells, and the model would be implicitly trained to fit those at the expense of the rest.

Together these operations produce $z_t \in \mathbb{R}^{11 \times 10}$, the surface S_t of Section 3.1.4 in standardised log-IV space. The task is posed in z : models receive inputs in this space, produce predictions in this space, and are scored against targets in this space.

3.2.3 Forecast Horizons

Four horizons are used: $h \in \{10, 21, 42, 63\}$ trading days, approximately two weeks, one month, two months, and one quarter ahead. The range is chosen to span the window over which the dynamics literature reviewed in Section 2.2.2 identifies the surface as partially predictable.

Forecast windows are formed by sliding a base date t through the calendar. Each window’s forecast path is the h -step trajectory $z_{t+1}, \dots, z_{t+h}$. A window is assigned to a split if and only if its forecast path lies entirely within that split’s date range.

3.3 Classical Benchmarks

Two classical benchmarks are reported in the results: persistence and the polynomial-coefficient vector autoregression of Gonçalves and Guidolin [7]. Persistence is the trivial forecast that the surface stays where it is; the polynomial-coefficient Vector Auto-regression (VAR) is the canonical reduced-form approach, reducing each daily surface to five coefficients and forecasting their dynamics by a linear model.

3.3.1 Persistence

The persistence forecast at base date t is the most recently observed surface, held constant across the forecast path out to each horizon:

$$\hat{z}_{t+i} = z_t \quad \text{for } i = 1, \dots, h, \quad h \in \{10, 21, 42, 63\}, \quad (3.4)$$

where z_t denotes the standardised log-IV surface. Persistence has no parameters, requires no training, and produces a flat path of predictions identical at every step.

Persistence is the optimal forecast under the assumption that the standardised log-IV surface is a martingale — that is, that the conditional expectation of the future surface given all past surfaces equals today's:

$$\mathbb{E}[z_{t+h} \mid z_t, z_{t-1}, \dots] = z_t.$$

Under this assumption no information in past surfaces is useful for forecasting future surfaces.

The dynamics literature reviewed in Section 2.2.2 characterises the surface as close to a martingale at shortest horizons: the near-unit-root autocorrelation structure documented by Gospodinov et al. [6] implies that today's surface explains a large share of the variance of the surface a few days out, so persistence is hard to beat over short horizons. Empirical studies have found persistence competitive with parametric forecasting models in this setting [7], [10]. Persistence is therefore the right null: the surface is close enough to a martingale that it is hard to beat at short horizons, but the small departure from it is exactly what the trained models are built to capture.

3.3.2 The Polynomial-Coefficient VAR

The model has two stages: a daily cross-sectional fit that reduces each surface to five coefficients, and a vector autoregression that forecasts those coefficients forward.

Stage one: cross-sectional reduction

On each day t , the gridded log-IV surface is fitted, by Ordinary Least Squares (OLS) across all 110 cells, to a polynomial in moneyness and maturity:

$$\ell_t(m, \tau) = \beta_{0,t} + \beta_{1,t}M + \beta_{2,t}M^2 + \beta_{3,t}\tau + \beta_{4,t}(M\tau) + \varepsilon, \quad M \equiv \frac{m}{\sqrt{\tau}}, \quad (3.5)$$

where $\ell_t = \log \sigma_t$ and M is the time-adjusted moneyness of Gonçalves and Guidolin, which rescales log-moneyness by $\sqrt{\tau}$ so that a given value of M corresponds to comparable moneyness across maturities.⁷ Each day is thereby reduced to a five-vector $\beta_t = (\beta_{0,t}, \dots, \beta_{4,t})$, the time series of which carries the surface's dynamics.

Five coefficients capture the surface closely: the per-day cross-sectional R^2 of the fit averages 0.962 over the sample, and exceeds 0.94 even in the worst-fitting decile of days.

Stage two: vector autoregression

The coefficient series is modelled by a vector autoregression,

$$\beta_t = c + \sum_{i=1}^p \Phi_i \beta_{t-i} + u_t, \quad (3.6)$$

fitted on the training set only, with lag order p chosen by the Bayesian Information Criterion (BIC) over $p \in \{1, \dots, 12\}$ and the coefficient matrices Φ_i estimated by OLS equation-by-equation. OLS targets one-step prediction MSE in the β space. BIC selects $p = 2$, and the fitted parameters are frozen thereafter.

A forecast at base date t is produced by iterating the frozen VAR forward step-by-step from $\hat{\beta}_{t+1}$ to $\hat{\beta}_{t+h}$, then mapping each forecast back to a surface through the Stage-one polynomial at every grid point. The result is a path of surface forecasts $\hat{z}_{t+1}, \dots, \hat{z}_{t+h}$.

⁷This is the specification of Gonçalves and Guidolin [7]. The original is fitted to individual option quotes each day; here it is fitted to the constructed, gridded surface.

3.4 Model Implementations

Every model in this section is built on one forecasting contract — the same input window, centring, and target — and is fitted under the same procedure. What differs is the predictor placed on top of that contract.

The transformer architectures form a ladder, from vanilla transformers at one end to the full Surface-Aware Neural Tensor Attention (SANTA) model at the other, with controlled ablations between them. What varies along the ladder is how each architecture consumes the surface: how the lookback is tokenised, what coordinate information is supplied, and how attention mixes across cells. Within the ladder every element other than the architecture is shared — each transformer uses the same hyperparameters at roughly the same parameter count ($\approx 40 - 50\text{k}$), with no tuning.

Beneath the ladder sits a linear floor: the minimal predictor the same contract admits, carrying neither attention nor the ladder’s capacity.

Input window, centring, and target

Implied-volatility surfaces evolve slowly in level, so all models introduced in this section forecast deviations from today’s surface rather than the absolute surface itself — concentrating capacity on the dynamics rather than the level.

The lookback is fixed at $L = 63$ standardised log-IV surfaces ending on the base date t — roughly a calendar quarter. The window is centred on today: with $z_{t-L+1:t}$ the lookback, the centred input is

$$u_{t-L+1:t} = z_{t-L+1:t} - z_t, \quad (3.7)$$

so the last slice u_t is identically zero and every earlier slice is a move leading up to today. The input the model sees is therefore the recent dynamics from which today’s surface emerged, not the surfaces themselves.

The same anchor is used at the output. For horizon h , the model produces an h -step path of residual surfaces $\delta_1, \dots, \delta_h$, and the predicted surface at step i is

$$\hat{z}_{t+i} = z_t + \delta_i, \quad i = 1, \dots, h. \quad (3.8)$$

Predicting $\delta_i = 0$ corresponds exactly to persistence. At initialisation the residual head produces

only small perturbations around persistence, and training drives the residuals toward whatever deviation the data supports.

Centring removes today’s level from the encoder’s input, so the encoder sees only the recent dynamics. To make regime-aware forecasts, each model’s head receives two additional signals: today’s surface z_t and the per-cell standard deviation $\tilde{s}$ over the lookback window. These let the predicted residual respond to the current level and variability of the surface.

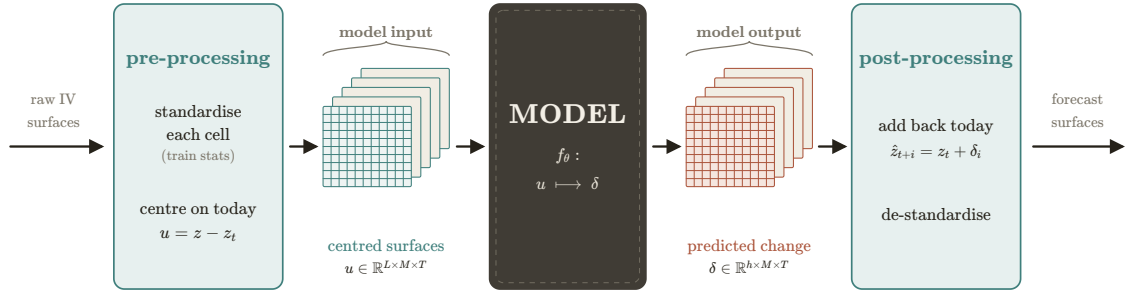


Figure 3.6: End-to-end pipeline shared by every model. Each model f_θ produces a path of residual surfaces δ , which is added back to today’s surface and de-standardised to give the forecast surfaces.

Windows near a split boundary have lookbacks that reach back into the prior split. These are past surfaces the model reads; the training forecast windows remain wholly within training, since each forecast window is assigned to the split containing its entire forecast path. (Section 3.2.3)

Training procedure

Each (architecture, horizon) pair is its own training run, fitted on the training set with model selection against the validation set. The training loss is the mean squared error of the predicted path against its realised counterpart in standardised log-IV space,

$$\mathcal{L}(t, h) = \frac{1}{h \cdot 110} \sum_{i=1}^h \|\hat{z}_{t+i} - z_{t+i}\|_F^2, \quad (3.9)$$

averaged over the h path-steps and the 110 surface cells, then over mini-batches of 64 windows during stochastic optimisation.

Optimisation is by AdamW [29] with learning rate 5×10^{-4} , weight decay 10^{-3} , and gradients clipped to a maximum norm of 1.0. Dropout of 0.1 is applied inside attention and feed-forward blocks and on the input tokens. The variations along the ladder concern attention and tokenisation rather than optimisation-relevant structure, so this configuration is held identical for every

transformer model and every horizon, with no per-model tuning.

Each run uses an NVIDIA A100 GPU via Google Colab. Training runs for up to 100 epochs with a patience of 15 epochs. The model state with the lowest validation loss is restored before test-set evaluation. Ten random seeds are used per training run;⁸ results are reported as the mean and standard deviation across seeds.

Per-model architectural details — embedding dimension, number of layers, number of attention heads, and total parameter counts — are given in Appendix B.

The Linear Floor (NLinear)

Before introducing the attention machinery shared by the transformer architectures of the ladder, a purely linear predictor is introduced as a learned baseline. Linear models are not a courtesy baseline in multivariate forecasting. Zeng et al. [30] showed that single-layer linear predictors matched or surpassed the leading forecasting transformers of the time across a range of long-horizon multivariate benchmarks.

Each cell’s lookback is centred on its last observed value, and the network predicts a path of deltas $\{\delta_{t+i}\}_{i=1}^h$ added back through the persistence reconstruction $\hat{z}_{t+i} = z_t + \delta_{t+i}$. Subtracting the last value, applying one linear layer, and adding it back is exactly the *NLinear* model of Zeng et al. [30]; the centring used in this experiment therefore makes the linear floor an NLinear model. Concretely, for surface cell c at target horizon h , NLinear learns a single weight matrix $W \in \mathbb{R}^{h \times L}$, shared across every cell of the grid, whose i -th row w_i produces the forecast at step i ,

$$\hat{z}_{t+i}^{(c)} = z_t^{(c)} + w_i^\top \left(z_{t-L+1:t}^{(c)} - z_t^{(c)} \mathbf{1} \right), \quad i = 1, \dots, h, \quad (3.10)$$

where the bracketed term is the persistence-centred lookback and the leading $z_t^{(c)}$ restores the level.

The defining property of this predictor is what it cannot do: the linear map is static, applying the same weighting of recent lags regardless of the date or the prevailing market state. It has no mechanism by which the response to the recent path could differ between a calm regime and a stressed one. The following transformer architectures, by contrast, can in principle condition their mixing of information on the current context. Reading the ladder against this floor therefore isolates a single question: whether the regime-conditional, context-dependent mixing that attention provides earns anything beyond a fixed linear projection.

⁸The integers 0 through 9, set via `torch.manual_seed`.

The training recipe is largely identical — lookback $L = 63$, batch size 64, the same early-stopping schedule (minimum 15, maximum 100 epochs, patience 15), the same optimiser (AdamW), the same hardware, and the same ten seeds per horizon — with two exceptions: the learning rate is raised to 4×10^{-3} and the weight decay is set to 0. These reflect what a single-layer, convex least-squares objective requires rather than what the deep models use — there is no excess capacity for weight decay to regularise, and the larger step suits the simpler objective.

Common architectural primitives

All five transformer architectures share the same internal machinery: a transformer encoder block [24] built from multi-head self-attention and a feed-forward network, with a lag embedding to supply position along the time axis.

The transformer block. A transformer block takes a sequence of vectors — one per input token — and produces a sequence of the same length, with each output now informed by the others. It does this in two stages: a multi-head self-attention operation that mixes information across tokens, then a feed-forward network that processes each token independently. Both stages run in pre-layer-norm residual form,

$$x \leftarrow x + \text{MHA}(\text{LN}(x)), \quad x \leftarrow x + \text{FFN}(\text{LN}(x)), \quad (3.11)$$

so each operation adjusts a maintained representation rather than overwriting it; the residual additions also keep gradients flowing cleanly when blocks are stacked into an encoder.

Multi-head self-attention. Self-attention lets each token incorporate information from every other token in the sequence. Three linear projections produce, for each token, a query, a key, and a value. The dot product of one token’s query with each key gives a score; the softmax of those scores forms the weights with which the corresponding values are averaged:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \quad (3.12)$$

where Q, K, V are the stacked queries, keys, and values across the sequence, d_k is the per-head key dimension, and the $\sqrt{d_k}$ scaling keeps the softmax from saturating at large d_k .

One such operation captures one pattern of who reads from whom. *Multi-head* attention runs

several in parallel, each with its own learned projections, and concatenates their outputs through a final linear map:

$$\text{MHA}(X) = W_O \text{concat}(\text{head}_1, \dots, \text{head}_{n_{\text{heads}}}), \quad \text{head}_h = \text{Attention}(XW_h^Q, XW_h^K, XW_h^V). \quad (3.13)$$

The heads can specialise: one might focus on the most recent steps, another on slower regime structure.

Feed-forward network. After attention, each token passes independently through a small feed-forward network: two linear layers with a GELU nonlinearity⁹ between them,

$$\text{FFN}(x) = W_2 \text{GELU}(W_1 x + b_1) + b_2. \quad (3.14)$$

Where attention moves information *between* tokens, the FFN processes information *within* each token.

Lag embedding. Attention as just described is order-agnostic in its inputs: reshuffling the tokens would reshuffle the output the same way and otherwise leave it unchanged. When attention runs along the time axis, lag is supplied explicitly by a learned table of L vectors $e_l \in \mathbb{R}^d$, one per lag, added element-wise to the token at lag l :

$$x_l \leftarrow x_l + e_l. \quad (3.15)$$

The table is learned jointly with the rest of the model.

A final layer-norm is applied after the encoder stack. The transformer model ladder uses $n_{\text{layers}} = 2$ encoder blocks and $n_{\text{heads}} = 4$ attention heads throughout; these are held fixed across the architectures so that the comparison rests on attention and tokenisation choices alone.

3.4.1 Day-Token Transformer

In the day-token transformer, a token represents a day. The centred surface $u_l \in \mathbb{R}^{M \times T}$ for day $l \in \{1, \dots, L\}$ is flattened to a vector of length $MT = 110$ and projected to the embedding

⁹GELU is the Gaussian Error Linear Unit, $\text{GELU}(x) = x \cdot \Phi(x)$ where Φ is the standard normal CDF; introduced by Hendrycks and Gimpel [31] as a smooth alternative to ReLU.

dimension by a single linear map,

$$x_l = W_{\text{in}} \text{vec}(u_l) + b_{\text{in}}, \quad W_{\text{in}} \in \mathbb{R}^{d \times MT}, \quad (3.16)$$

and the lag embedding of equation (3.15) is added to give the day’s input token $x_l + e_l$. The encoder then attends over the $L = 63$ day-tokens. This model has no explicit knowledge of cell geometry — every cross-cell relationship it can use is mediated by W_{in} at the input.

After the encoder stack, the day- L token is taken as the model’s summary of recent dynamics,

$$r = x_L \in \mathbb{R}^d. \quad (3.17)$$

Today’s surface z_t and the per-cell standard deviation $\tilde{s}$ are each flattened, projected to d , and added to r :

$$r \leftarrow r + W_{\text{lvl}} \text{vec}(z_t) + W_{\text{scl}} \text{vec}(\tilde{s}). \quad (3.18)$$

A single linear head then expands the d -vector to the full residual path,

$$\text{vec}(\delta_1, \dots, \delta_h) = W_{\text{out}} r, \quad W_{\text{out}} \in \mathbb{R}^{(MT \cdot h) \times d}, \quad (3.19)$$

reshaped to give the residual path.

The output head is the structural cost of the day-token design. Every cell, at every horizon step, is reconstructed from the same d -vector, and W_{out} alone carries $MT \cdot h \cdot d$ parameters that scale linearly with the forecast horizon. To keep the total parameter count comparable to the rest of the ladder ($\approx 40 - 50\text{k}$), d is reduced as h grows: $d = 24, 16, 8, 8$ for $h = 10, 21, 42, 63$ respectively. At the longest horizon, even with d at 8, the output head dominates and the model lands at roughly 66k parameters. This excess over the target is accepted, since the day-token design sits at the floor of the transformer ladder.

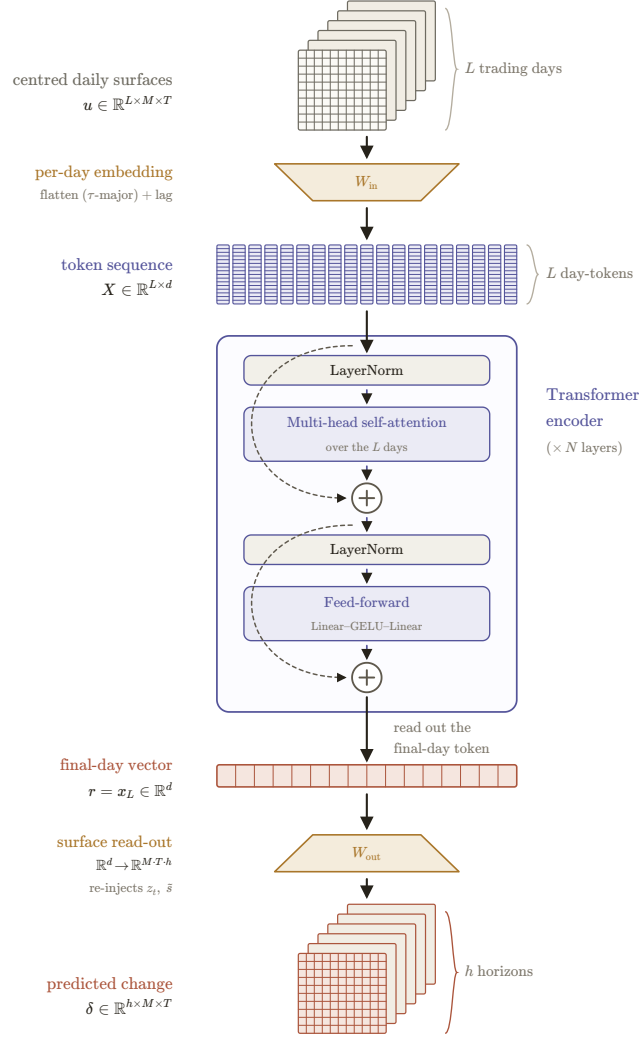


Figure 3.7: Day-Token Transformer. Each day’s surface is flattened into a single token; the encoder attends across day-tokens; a surface-wide output head reconstructs the full residual path.

3.4.2 Per-Cell Transformer

In the per-cell transformer, a token represents a single cell on a single day. Each surface cell carries its own token sequence of length L , attention runs along the temporal axis of every cell independently, and the model has no information about cell geometry. Unlike the day-token transformer, each cell preserves its own representation through the encoder stack rather than being collapsed through the day-projection.

For each cell (m, τ) and each day l in the window, the centred scalar $u_{l,m,\tau}$ is projected to a

d -dimensional token by a single linear map:

$$x_{l,m,\tau} = W_{\text{val}} u_{l,m,\tau} + b_{\text{val}}, \quad W_{\text{val}} \in \mathbb{R}^{d \times 1}, \quad (3.20)$$

and the lag embedding of equation (3.15) is added independently to each cell’s sequence. The same value projection is used for every cell, and the result is a tensor of tokens $x \in \mathbb{R}^{L \times M \times T \times d}$, a factor of $MT = 110$ more tokens than the day-token model uses to represent the same window.

The transformer block of equation (3.11) is applied to the temporal sequence of each cell separately, with the same attention and feed-forward weights shared across all cells. Cells may differ in level — a near-ATM short-dated cell behaves very differently from a deep-OTM long-dated one — but the structure of how they evolve in time is assumed to be shared, and weight-sharing makes this assumption explicit while keeping the model parameter-efficient.

The output head is per-cell rather than surface-wide. After the encoder stack, the day- L representation $r_{m,\tau} = x_{L,m,\tau}$ is each cell’s own summary of recent dynamics. A shared two-layer MLP produces the residual path at that cell from $r_{m,\tau}$ augmented with $z_t(m, \tau)$ and $\tilde{s}(m, \tau)$:

$$(\delta_1, \dots, \delta_h)(m, \tau) = \text{MLP}([r_{m,\tau}; z_t(m, \tau); \tilde{s}(m, \tau)]). \quad (3.21)$$

The bottleneck of the day-token output head is gone — and with it the parameter cost it carried, so d is fixed across horizons at $d = 56$.

Together, per-cell tokenisation and weight sharing across cells mean the model can capture rich temporal patterns within each cell but cannot represent cross-cell dynamics at all. Whether a quote at $m = -0.05$ leads or lags a quote at $m = 0$ is something the per-cell transformer cannot express.

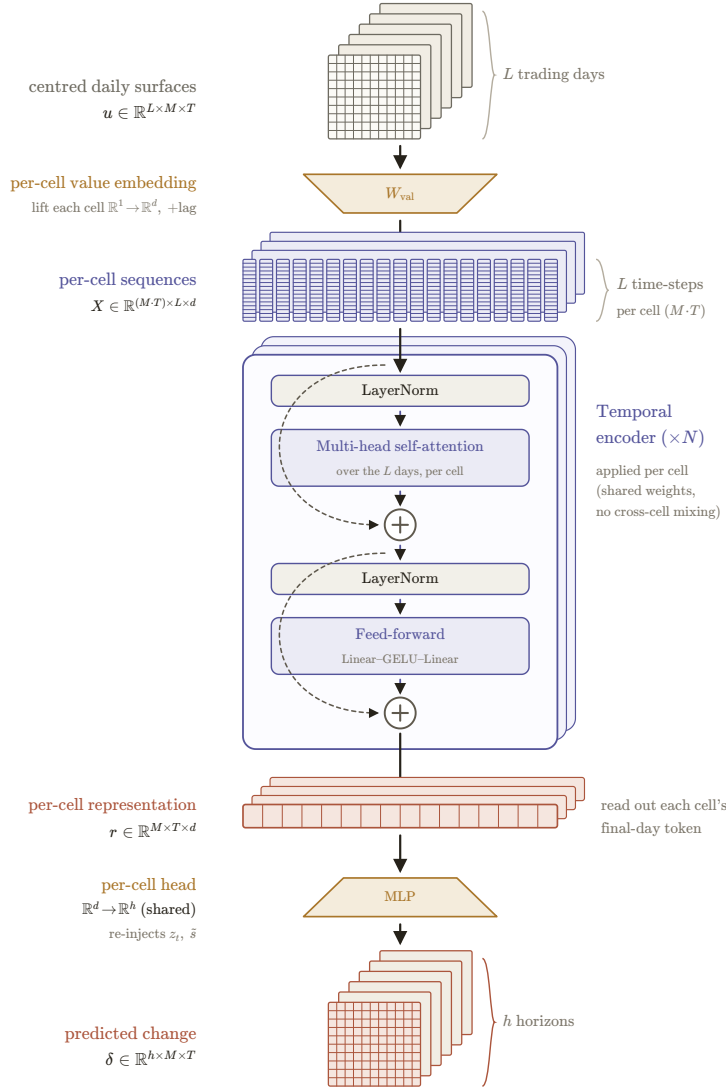


Figure 3.8: Per-Cell Transformer. Each surface cell carries its own length- L token sequence; the transformer block is applied to each cell’s sequence in parallel with shared weights, and the per-cell head produces each cell’s residual path independently.

3.4.3 Surface-Aware Embeddings

The day-token and per-cell transformers had no information about where on the surface each cell sits. The remaining models in the ladder — the SANTA family — add that information through *coordinate embeddings*: a learned map from each cell’s financial coordinate to a d -dimensional embedding vector. This component is shared identically across the three SANTA architectures.

Two coordinate axes are embedded: log-forward-moneyness m and log-maturity $\log \tau$. Each axis is mapped through its own two-layer MLP with a GELU nonlinearity,

$$\text{Embed}_{\text{coord}}(c) = W_2 \text{GELU}(W_1 c + b_1) + b_2, \quad W_1 \in \mathbb{R}^{d \times 1}, W_2 \in \mathbb{R}^{d \times d}. \quad (3.22)$$

Embedding the two axes separately lets skew and term-structure dynamics be learned independently.

The coordinate embeddings are added to the per-cell value embedding (equation 3.20) and the lag embedding (equation 3.15), both unchanged from the per-cell transformer, to form each cell-day's token:

$$x_{l,m,\tau} = W_{\text{val}} u_{l,m,\tau} + \text{Embed}_m(m) + \text{Embed}_\tau(\log \tau) + e_l. \quad (3.23)$$

All four live in the same d -dimensional residual stream from the start, and the model learns their relative weighting internally.

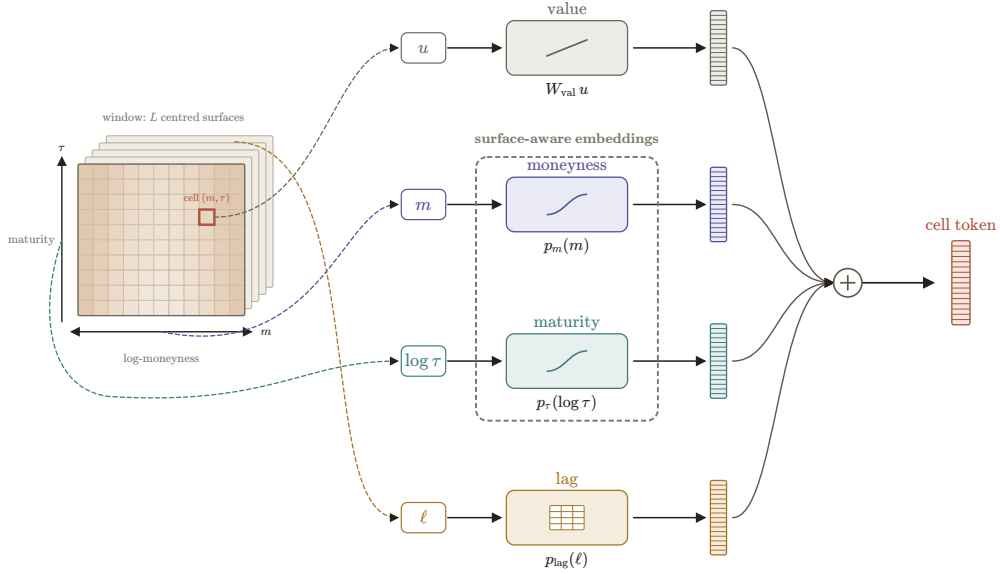


Figure 3.9: Surface-aware token assembly. Coordinate embeddings for moneyness and maturity are summed with the per-cell value and lag embeddings into the token.

3.4.4 SANTA-Temporal

SANTA-Temporal is the per-cell transformer of Section 3.4.2 with the coordinate embeddings of Section 3.4.3 added to each cell-day token. Tokenisation, backbone, and head are otherwise unchanged: each cell carries its own length- L sequence, attention runs per cell along the temporal axis with weights shared across cells, and the per-cell head reconstructs each cell's residual path independently. The embedding dimension is fixed across horizons at $d = 56$.

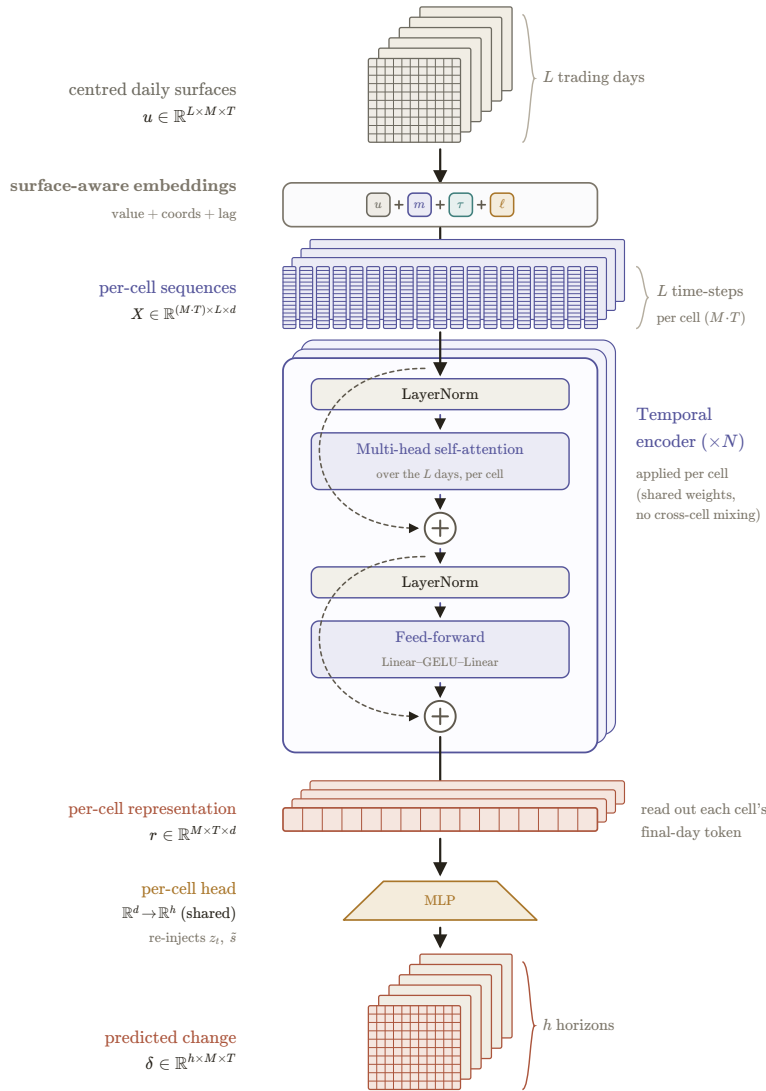


Figure 3.10: SANTA-Temporal. Each surface cell carries its own length- L token sequence, with surface-aware embeddings added to each token; the encoder attends along the temporal axis of every cell with shared weights.

The coordinate embeddings let the model relate cells of similar coordinates implicitly: the same weights firing on temporally similar histories at nearby coordinates produce similar outputs. But no information ever flows between cells inside the model. A move at $m = -0.05$ on day t is never read by the cell at $m = 0$ on day t , no matter how strongly the two coordinates suggest they should be connected.

3.4.5 SANTA-Flat

SANTA-Flat adds a spatial attention sub-block to the SANTA-Temporal encoder layer. Each cell can now read from every other cell of the same day, not just from its own history. Tokenisation, coordinate embeddings, and the per-cell head are unchanged from Section 3.4.4; the embedding dimension is reduced to $d = 40$ to keep the total parameter count within $\approx 40 - 50\text{k}$ after the addition of the new sub-block.

The spatial attention treats each day’s surface as a flat sequence of $MT = 110$ cells. For each day in the window, the day’s $M \times T \times d$ slice of tokens is flattened to a sequence of length MT and passed through multi-head self-attention as in equation (3.13); the output is reshaped back. Each cell’s representation is now updated, at each layer, by a learned weighted average over all 110 cells of its own day — the model’s first capability to route information across the surface.

Each encoder layer applies three sub-operations: spatial attention across cells of the same day, temporal attention along each cell’s history, and the position-wise FFN, all in the pre-LN residual form of equation (3.11):

$$\begin{aligned} x &\leftarrow x + \text{SpAttn}(\text{LN}(x)), \\ x &\leftarrow x + \text{TempAttn}(\text{LN}(x)), \\ x &\leftarrow x + \text{FFN}(\text{LN}(x)). \end{aligned} \tag{3.24}$$

Spatial attention runs first, so the temporal attention reads tokens that have already been mixed across the surface.

The “flat” in the name records what this architecture does not yet do: the 110 cells are treated as a flat sequence, with no architectural recognition that the surface is a tensor with two distinct axes — moneyness and maturity. The full-attention spatial sub-block has $O((MT)^2)$ pairwise interactions per day per head.

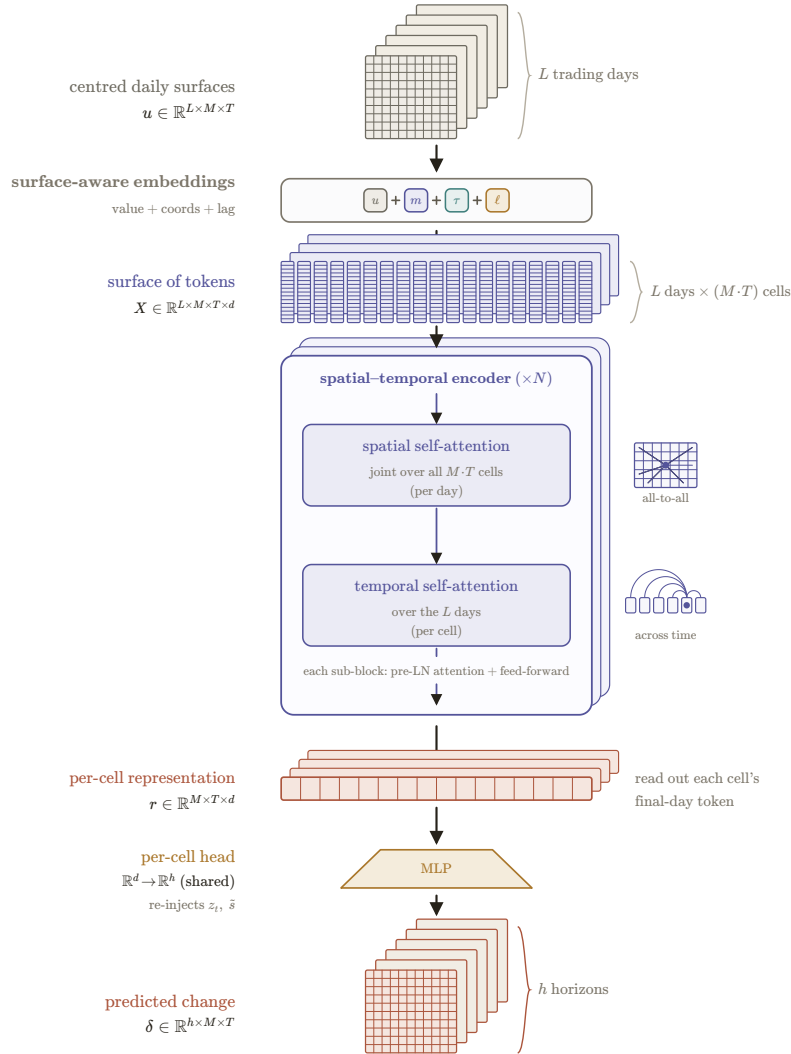


Figure 3.11: SANTA-Flat. Per-cell tokenisation with coordinate embeddings; each encoder layer applies spatial attention across all 110 cells of a day, then temporal attention along each cell’s history.

3.4.6 SANTA

SANTA — Surface-Aware Neural Tensor-Attention — replaces the joint spatial attention of SANTA-Flat with two axis-aligned spatial sub-blocks: one over the moneyness axis, one over the maturity axis. Tokenisation, coordinate embeddings, the temporal sub-block, and the per-cell head are unchanged from Section 3.4.5; the embedding dimension is reduced to $d = 32$ to keep the total parameter count $\approx 40 - 50\text{k}$ after the added sub-block.

The factored attention operates on slices of the day’s token tensor. The moneyiness attention treats each fixed-maturity column as a sequence of M tokens; the maturity attention treats each fixed-moneyiness row as a sequence of T tokens. Both use the multi-head self-attention of equation (3.13); they differ only in which axis is the sequence:

$$\begin{aligned}\text{MoneyAttn}(X_{l, :, \tau, :}) &= \text{MHA}(X_{l, :, \tau, :}) \quad \text{for each } \tau, \\ \text{MatAttn}(X_{l, m, :, :}) &= \text{MHA}(X_{l, m, :, :}) \quad \text{for each } m,\end{aligned}\tag{3.25}$$

with the multi-head weights shared across the orthogonal axis in each case.

Each encoder layer now applies four sub-operations: moneyiness attention, maturity attention, temporal attention, and the FFN, all in the pre-LN residual form of equation (3.11):

$$\begin{aligned}x &\leftarrow x + \text{MoneyAttn}(\text{LN}(x)), \\ x &\leftarrow x + \text{MatAttn}(\text{LN}(x)), \\ x &\leftarrow x + \text{TempAttn}(\text{LN}(x)), \\ x &\leftarrow x + \text{FFN}(\text{LN}(x)).\end{aligned}\tag{3.26}$$

The two spatial sub-blocks run before the temporal one, so the temporal attention reads tokens already mixed across both surface axes.

This factoring — attending along one axis of the surface at a time rather than over all cells jointly — is the axial attention of Ho et al. [32], here applied to the two spatial axes of the volatility surface; running the spatial axes separately from, and ahead of, the temporal one follows the divided space–time attention of Bertasius et al. [33]. SANTA therefore never forms the joint attention over all MT cells that SANTA-Flat computes.

Factored attention, as implemented here, mixes the surface one axis at a time — first across moneyiness at each maturity, then across maturity at each moneyiness — rather than letting every cell attend to every other. It is thus a restricted, less flexible form of the joint attention SANTA-Flat uses, and the restriction cuts the per-day cost from the $O((MT)^2)$ of joint attention to $O(M^2) + O(T^2)$. The factoring builds in an assumption that skew and term-structure dynamics are separable axes of variation in the surface, and bets that this inductive bias outweighs the lost flexibility. If the assumption holds, the factored attention learns axis-specialised structure that SANTA-Flat must discover from scratch.

SANTA closes the ladder. The progression from the day-token transformer through the per-cell

transformer, SANTA-Temporal, SANTA-Flat, and finally SANTA is a sequence of single architectural additions: tokenisation per cell, coordinate awareness, cross-cell routing, and factoring along the surface's natural axes.

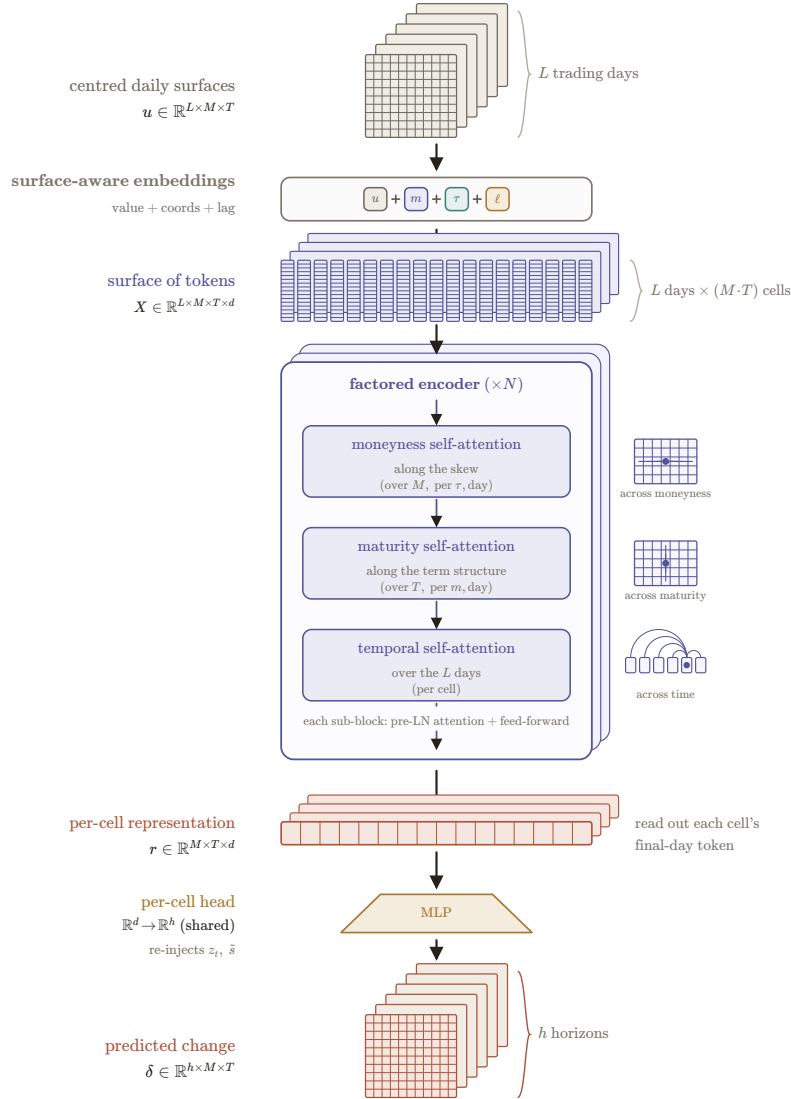


Figure 3.12: SANTA. Per-cell tokenisation with coordinate embeddings; each encoder layer applies moneyiness attention across strikes, maturity attention across maturities, and temporal attention along each cell's history.

3.5 Evaluation Criteria

Every model is scored on three metrics — Mean Squared Error (MSE), Mean Absolute Error (MAE), and Information Coefficient (IC) — each computed in standardised log-IV space and averaged over the full forecast path. For the seeded deep models each metric is computed per seed and reported as mean $\pm$ standard deviation over the ten seeds, while the deterministic baselines carry no standard deviation.

Point-wise error metrics

Letting $\hat{z}_{n,i,c}$ denote the prediction in test window n ,¹⁰ at horizon step $i = 1, \dots, h$ and surface cell $c = 1, \dots, 110$, and $z_{n,i,c}$ the realised value, the mean squared and mean absolute errors measure the size of the forecast error directly:

$$\text{MSE} = \frac{1}{110 N h} \sum_{n=1}^N \sum_{i=1}^h \sum_{c=1}^{110} (\hat{z}_{n,i,c} - z_{n,i,c})^2, \quad (3.27)$$

$$\text{MAE} = \frac{1}{110 N h} \sum_{n=1}^N \sum_{i=1}^h \sum_{c=1}^{110} |\hat{z}_{n,i,c} - z_{n,i,c}|. \quad (3.28)$$

Both are zero for a perfect forecast and grow with the discrepancy; lower is better. MSE is the training objective of Equation (3.9), so the models are fitted to minimise exactly this quantity. MAE penalises the same errors linearly rather than quadratically and is reported alongside as a robustness check, being less sensitive to a small number of large misses.

Cross-sectional correlation metric

The IC measures whether the model ranks the cells' moves correctly across the surface, regardless of how large those moves are. Let $x_{n,c}$ be today's value at cell c — the last day of the lookback. For each window and step, IC takes the predicted and realised change relative to today,

$$\Delta \hat{z}_{n,i,c} = \hat{z}_{n,i,c} - x_{n,c}, \quad \Delta z_{n,i,c} = z_{n,i,c} - x_{n,c}, \quad (3.29)$$

and ranks the 110 cells by each:

$$\hat{R}_{n,i,c} = \text{rank}_c(\Delta \hat{z}_{n,i,\cdot}), \quad R_{n,i,c} = \text{rank}_c(\Delta z_{n,i,\cdot}). \quad (3.30)$$

¹⁰The number of test windows N shrinks slightly as h grows, since each forecast path must fit inside the test set.

The IC is the cross-sectional correlation of these two rankings — Spearman’s ρ between predicted and realised cell changes — averaged over every window and step:

$$\text{IC}_{n,i} = \frac{\sum_c (\hat{R}_{n,i,c} - \bar{R})(R_{n,i,c} - \bar{R})}{\sqrt{\sum_c (\hat{R}_{n,i,c} - \bar{R})^2} \sqrt{\sum_c (R_{n,i,c} - \bar{R})^2}}, \quad \text{IC} = \frac{1}{Nh} \sum_{n=1}^N \sum_{i=1}^h \text{IC}_{n,i}, \quad (3.31)$$

where $\bar{R}$ is the mean rank. The coefficient lies in $[-1, 1]$, higher being better. An IC of 1 at a given step means the model ordered all 110 cells’ moves exactly as they were realised — it perfectly predicted which parts of the surface would rise or fall most relative to today; an IC of 0 matches a random ordering, and -1 reverses it.

Because it reads only the order of the cell changes and not their size, the IC is unchanged by a level shift of the whole surface or a common positive rescaling of the cell moves — either leaves the ranking, and hence the IC, untouched. It is therefore robust to the prevailing level of the SPX surface.

Persistence has no IC: by setting $\hat{z}_{n,i,c} = x_{n,c}$ it predicts zero change at every cell and the rank correlation is undefined.

4

Results

Contents

4.1 Aggregate Forecasting Accuracy	47
4.2 Point-wise Error	47
4.2.1 Post-crisis Period	49
4.2.2 Crisis Period	51
4.3 Cross-sectional skill (IC)	53
4.4 The Cost of Unconstrained Spatial Attention	56
4.5 Persistence as an Anchor	57
4.6 Summary of Results	58

This chapter compares the forecasting performance of the models and classical benchmarks over the test set. The forecast surfaces themselves are too numerous to reproduce here — multiple seeds, horizon steps, and regimes — so this chapter reports aggregate trends, with representative forecast surfaces, on the raw implied-volatility scale, collected in Appendix C.1. Before any of these comparisons, it is worth being precise about what the evaluation metrics actually measure, because the distinction between them governs how every table and figure below should be read.

The two point-wise metrics, MSE and MAE, are summed over all 110 cells of the surface. That sum is dominated by the surface’s overall *level* — by how well a model forecasts where the whole surface sits — and only weakly sensitive to its *shape*. The reason is the correlation structure of the surface’s movements. Section 2.2.2 established that the surface’s daily changes are concentrated in a few common modes, with an overall shift in level carrying $> 90\%$ of the variance. The per-cell standardisation rescales each cell but does not remove this common mode, so a level shift continues to dominate the summed loss in the standardised space the models are scored in. Because that common movement is shared across the surface, a level error lands on all 110 cells with the same

sign and adds across the grid, whereas a shape error spreads mixed-sign errors across different cells, each contributing less to the summed square than a coherent level error of the same total size. A loss summed over correlated cells therefore rewards getting the common level right far more than it rewards getting the shape right.

The IC is the complementary view. By ranking the cells’ moves and correlating predicted against realised order (Section 3.5), it is invariant to a monotone shift of all cells, measuring precisely the shape that the point-wise metrics largely overlook. The two families of metric thus separate cleanly: MSE and MAE mostly score the level, the IC scores the shape.

These results are reported throughout in the standardised log-IV space in which the models are trained. Reporting raw implied volatility error instead would re-introduce exactly the cross-cell scale differences that standardisation removes, weighting the comparison toward the most volatile cells for reasons unrelated to which model forecasts better.

4.1 Aggregate Forecasting Accuracy

Forecasting accuracy is first compared across the full test set (31 December 2019 to 29 December 2023). The comparison spans the five transformer architectures of the ladder — Day-Token, Per-Cell, SANTA-Temporal, SANTA-Flat, and SANTA — together with NLinear and the two classical baselines: persistence and the polynomial-coefficient VAR. Each is scored on MSE, MAE, and IC across paths spanning all four horizons $h \in \{10, 21, 42, 63\}$.

Table 4.1 reveals two different pictures depending on the metric. On MSE and MAE the trained models are tightly bunched: at every horizon the spread across the five transformer architectures and NLinear is small beside the gap separating all of them from persistence and VAR, with SANTA holding a narrow lead that widens into the longer horizons. On IC the ordering is far more separated — the SANTA family of models sit clear of the rest, followed by VAR and Per-Cell, while Day-Token is the only model whose IC does not increase with forecasting horizon.

4.2 Point-wise Error

The test set spans from 31 December 2019 to 29 December 2023, a window that opens with the COVID-19 crisis and goes on to cover a range of subsequent regimes. Because these episodes

Horizon	Model	MSE	MAE	IC
$h+10$	Persistence	0.10879	0.21160	—
	VAR	0.11919	0.22285	0.1273
	NLinear	0.11139 ± 0.0020	0.21215 ± 0.0018	0.0984 ± 0.019
	Day-Token	0.10898 ± 0.0006	0.20688 ± 0.0010	0.0945 ± 0.018
	Per-Cell	0.10766 ± 0.0008	0.20807 ± 0.0018	0.1119 ± 0.015
	SANTA-T	0.10687 ± 0.0007	0.20796 ± 0.0009	0.1474 ± 0.020
	SANTA-F	<u>0.10672 ± 0.0003</u>	<u>0.20820 ± 0.0010</u>	<u>0.1359 ± 0.014</u>
	SANTA	0.10641 ± 0.0004	<u>0.20726 ± 0.0006</u>	0.1292 ± 0.017
$h+21$	Persistence	0.22381	0.29419	—
	VAR	0.23307	0.29428	0.1646
	NLinear	0.22021 ± 0.0018	0.28799 ± 0.0017	0.1227 ± 0.009
	Day-Token	0.21769 ± 0.0013	0.28249 ± 0.0016	0.1029 ± 0.021
	Per-Cell	0.21160 ± 0.0015	0.28152 ± 0.0023	0.1603 ± 0.009
	SANTA-T	0.21066 ± 0.0036	0.28198 ± 0.0032	0.2178 ± 0.012
	SANTA-F	<u>0.21040 ± 0.0014</u>	<u>0.28130 ± 0.0016</u>	0.2167 ± 0.018
	SANTA	0.20972 ± 0.0011	0.28100 ± 0.0011	0.2247 ± 0.008
$h+42$	Persistence	0.41070	0.39585	—
	VAR	0.40941	0.38244	0.2164
	NLinear	0.38846 ± 0.0036	0.38005 ± 0.0026	0.1800 ± 0.011
	Day-Token	0.38832 ± 0.0020	0.37232 ± 0.0020	0.1061 ± 0.023
	Per-Cell	0.37168 ± 0.0033	0.36733 ± 0.0022	0.2276 ± 0.015
	SANTA-T	<u>0.36955 ± 0.0062</u>	<u>0.36690 ± 0.0047</u>	0.2748 ± 0.014
	SANTA-F	0.38458 ± 0.0219	0.37400 ± 0.0123	<u>0.2763 ± 0.020</u>
	SANTA	0.36487 ± 0.0030	0.36434 ± 0.0012	0.2905 ± 0.009
$h+63$	Persistence	0.50615	0.44730	—
	VAR	0.49163	0.42543	0.2530
	NLinear	0.47102 ± 0.0018	0.42909 ± 0.0010	0.2095 ± 0.004
	Day-Token	0.47243 ± 0.0046	0.41806 ± 0.0036	0.1058 ± 0.021
	Per-Cell	0.45652 ± 0.0200	0.41870 ± 0.0140	0.2348 ± 0.038
	SANTA-T	<u>0.44648 ± 0.0092</u>	<u>0.41201 ± 0.0064</u>	0.2865 ± 0.027
	SANTA-F	0.46303 ± 0.0369	0.41978 ± 0.0212	<u>0.3074 ± 0.019</u>
	SANTA	0.43882 ± 0.0059	0.40791 ± 0.0020	0.3158 ± 0.008

Table 4.1: Test-set accuracy across horizons (standardised log-IV), full test-set. Trained models reported as mean $\pm$ sd over 10 seeds. Best per horizon and metric in **bold**, second underlined.

differ so sharply in absolute implied volatility level, a single aggregate MSE or MAE need not be representative. It is therefore worth inspecting where, across the test period, the error is actually contributed.

The cumulative-share curve (Figure 4.1, bottom) shows the squared error jumping almost vertically at the onset of the COVID-19 crisis, which alone accounts for $\approx 60\%$ of the total MSE accumulated across the entire four-year test set, while the MAE curve climbs at the same dates but far more gently, reaching $\approx 25\%$.

The consequence is that the aggregate ranking of the models over the whole test set is largely determined by this single episode. The analysis is therefore split at 30 April 2020 into two complementary periods. The crisis window (31 December 2019 to 30 April 2020), and the post-crisis remainder (May 2020 to December 2023) containing the bulk of the test windows; this typical-

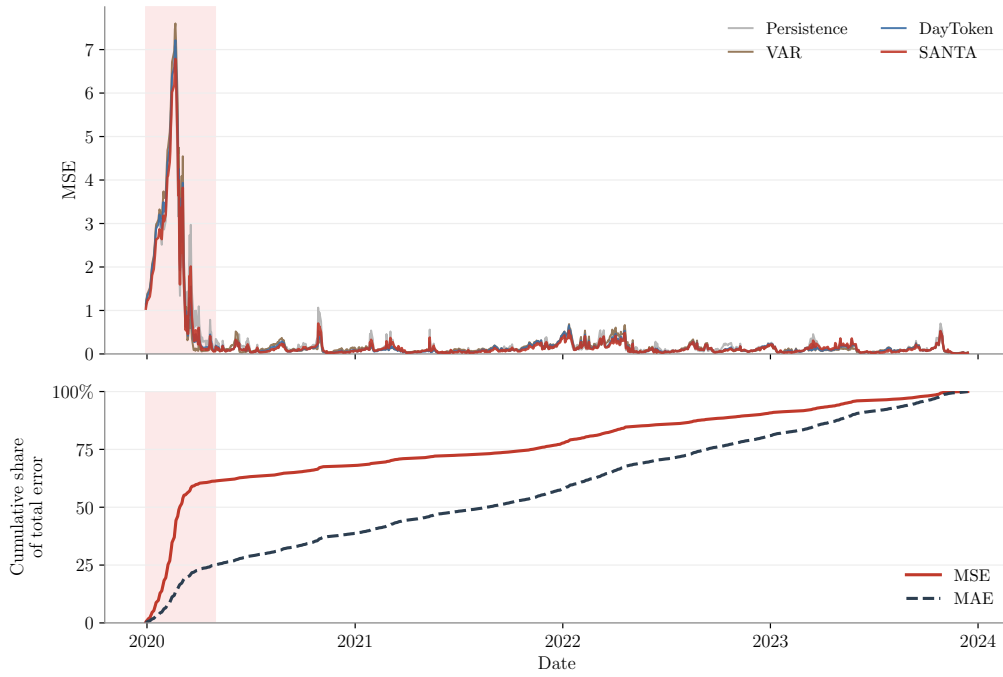


Figure 4.1: Concentration of forecast error over the test period. Top: per-window MSE, averaged over the four horizon paths and assigned to the path’s base date. Bottom: cumulative share of total MSE and MAE, pooled over all models, horizons, and windows.

regime stretch is examined first. Detailed tables, like the one above, limited to the crisis windows and the post-crisis remainder can be found in Appendix C.2.

4.2.1 Post-crisis Period

Restricted to May 2020 onward, Figure 4.2 shows forecast MSE over this period, by horizon and as a 63-day moving average through time.

The lower panel reports MSE by horizon, and shows the stringency of persistence as a benchmark. At $h+10$ the trained models cannot cleanly separate from it: the trained architectures span 0.061–0.063 against persistence’s 0.066, so holding today’s surface fixed is already a highly competitive forecast two trading weeks out, leaving little to improve on. As the horizon lengthens the level drifts from today and the flat forecast decays, so the trained models pull away from persistence as the path grows — by $h+63$ the best model reaches 0.168 against persistence’s 0.201.

At the short horizons the trained architectures are tightly bunched, sitting within a narrow band of one another on MSE: at $h+21$ they all fall between 0.099 and 0.104. From $h+42$ onward the ordering sharpens: persistence, VAR and NLinear show the largest errors (at $h+42$, 0.168, 0.160 and 0.156 respectively), followed by the transformers. At both $h+42$ and $h+63$, SANTA

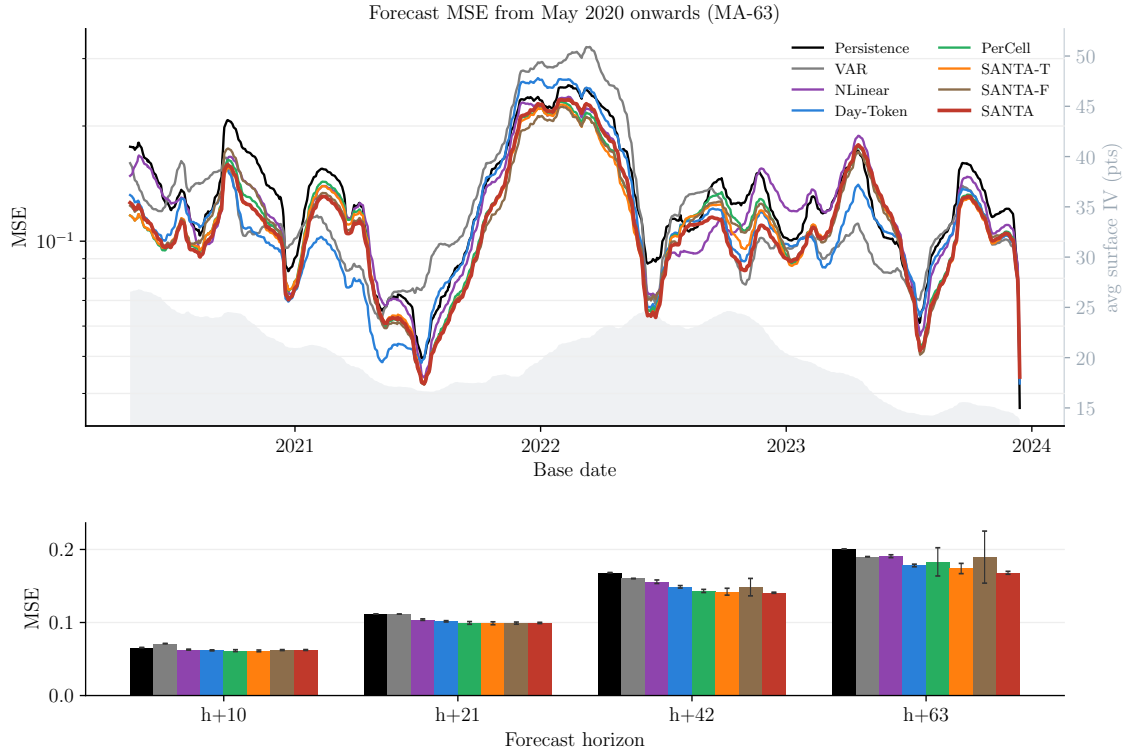


Figure 4.2: Forecast MSE over the post-crisis period (May 2020 onward), averaged across the ten seeds per model. Top: 63-day moving average through time, with each forecast path averaged over all four horizons and assigned to its base date. Bottom: MSE by horizon, with the seed standard deviation shown as whiskers.

achieves the lowest MSE followed by SANTA-Temporal (0.141 vs. 0.142 at $h+42$; 0.168 vs. 0.174 at $h+63$).

At the two longest horizon configurations, SANTA-Flat achieves worse MSE than the other two SANTA variants and Per-Cell (0.190 at $h+63$, against SANTA's 0.168, SANTA-Temporal's 0.174 and Per-Cell's 0.183), with comparatively very large seed standard deviations (± 0.036 at $h+63$ versus SANTA's ± 0.002 and SANTA-Temporal's ± 0.007).

The upper panel traces the 63-day moving average of MSE, averaged across all horizon paths and assigned to the forecast base date, with the average level of the SPX surface overlaid. The models' errors typically rise and fall together, finding the same periods difficult to forecast. The most prominent increase in MSE appears in the closing dates of 2021 — the base dates from which the models must forecast the rising implied volatility of the 2022 regime — where the horizon-averaged moving average more than doubles, peaking at roughly $2.2\times$ its 2023 calm-regime level.¹ Through the calmer stretches the SANTA variants and Per-Cell sit at the bottom of the band, but

¹Through 2022 the SPX surface sat at a sustained elevated level, as persistent inflation and the Federal Reserve's interest rate hikes drove the S&P 500 down roughly 19% over the year.

they cross one another frequently and no single model holds the outright lead on more than about a fifth of base dates. Reading the error evolution against the surface level reveals a pattern: at base dates followed by a sharp fall in the average level, the usual ranking (of the SANTA variants and Per-Cell leading) flips, and VAR and Day-Token — the two models that collapse the surface’s cross-section rather than representing it per cell — lead the field across the sharpest-falling base dates. When the level is rising, or consistently high, the same two produce the largest errors. VAR reverts toward the long-run level it was fitted on in training, so it is rewarded when the surface is falling toward that level and penalised when it rises away; Day-Token follows the same directional pattern. NLinear records higher error than all four per-cell transformers (Per-Cell and the SANTA variants) on roughly 72% of base dates, losing out through its inability to adapt its response to the regime each lookback reflects.

Over the post-crisis period, MAE reveals a similar picture. The models are only separated at the longer horizon configurations, at $h+63$ SANTA has the lowest MAE (0.320) followed by SANTA-Temporal (0.325); at the short horizons the field is bunched (e.g. all trained models within 0.180–0.186 at $h+10$). Across time, VAR and Day-Token produce the lowest errors when the average SPX surface level falls aggressively, with the transformers leading the field otherwise. The corresponding figure is deferred to Appendix C.3.

4.2.2 Crisis Period

The crisis window runs from the first test base date (31 December 2019) through 30 April 2020, covering the onset of the COVID-19 volatility spike. Figure 4.3 shows forecast MSE over this period, by horizon and as a 5-day moving average through time; the shorter averaging window reflects the small number of base dates the period contains.

The errors here dwarf those of the post-crisis period — larger by roughly nine-fold at $h+10$ and approaching twenty-fold at $h+63$ — and they should. The COVID-19 spike drove the SPX surface level upward faster and further than anything in the lookback could foreshadow. What the period does reveal is how the models differ in the parts of the episode following the prominent spike.

Through the run-up to the peak the SANTA family and Per-Cell carry the lowest error of the field, closely bunched and difficult to separate by eye; SANTA is the single most often lowest, leading on roughly 43% of the run-up base dates. This stretch matters out of proportion to its length, because it is where the absolute error is at its largest. SANTA’s narrow lead within the leading cluster here is, to a large degree, what places it first both on the full test-set aggregate

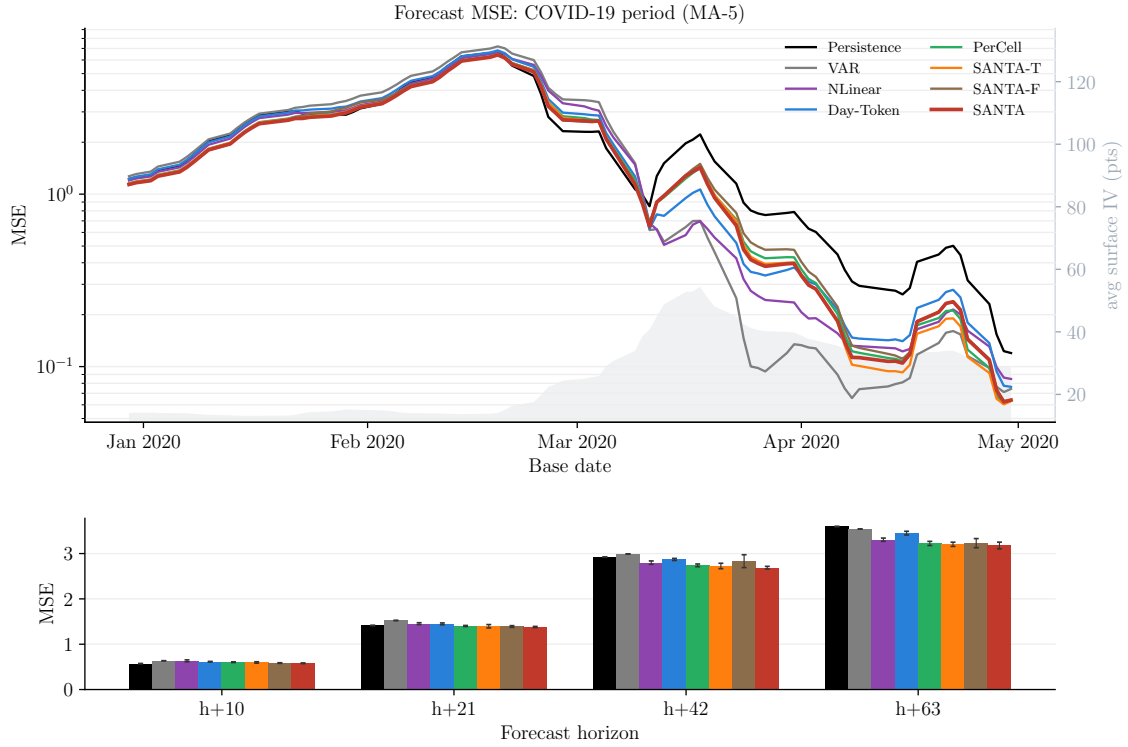


Figure 4.3: Forecast MSE over the COVID-19 crisis period, averaged across the ten seeds per model. Top: 5-day moving average through time, with each forecast path averaged over all four horizons and assigned to its base date, and the average level of the SPX surface overlaid. Bottom: MSE by horizon, with the seed standard deviation shown as whiskers.

table (Table 4.1) and in the by-horizon ranking here — where it is lowest at $h+21$, $h+42$ and $h+63$ by slim margins (e.g. 1.380 against SANTA-Flat’s 1.391 at $h+21$), persistence edging it only at $h+10$.

The ranking inverts on the way down. Once the level has peaked (mid-March) and begins to fall, VAR produces the lowest error, followed by NLinear and Day-Token: through the immediate recovery their smoothed MSE runs at roughly 0.23, 0.32 and 0.45 against about 0.53 for the SANTA variants. Again here, VAR reverts confidently toward the long-run level it was fitted on, and when the surface happens to collapse toward that level, confident reversion is exactly what is rewarded. Despite having been exposed to the 2008 crisis in training, the other transformer models hesitate to forecast a snap-back as violent as the one that occurred, and trail the simpler models through the immediate recovery.

For a stretch of base dates from late February into early March 2020 — with the absolute peak within the models’ forecast horizons — persistence is briefly the best forecast of the field. The moment the peak passes, the models’ downward predictions become correct and they separate

sharply, with persistence remaining last from there to the end of the window.

By the very end of the window, as the level stabilises, the SANTA variants retake the lead. The crisis period confirms the reversion pattern already seen post-crisis: confident mean-reversion — VAR above all — is what wins when the level is falling fast.

Over the crisis episode, MAE produces the same model rankings and qualitative picture. The corresponding figure is deferred to Appendix C.4.

4.3 Cross-sectional skill (IC)

Where MSE and MAE score higher errors with increasing horizons, the full-test aggregate table (Table 4.1) shows IC doing the opposite: it is weakest at $h+10$ and strengthens monotonically out to $h+63$ for every trained model.² This seems counter-intuitive at first — forecasts of more distant surfaces are usually worse, not better — but it follows from what the cross-sectional rank IC measures and how the surface’s shape evolves.

Prior work in Section 2.2.2 established that the surface’s shape mean-reverts: from any starting point it drifts back toward a long-run norm. Whether a ranking metric can detect this depends on how the predictable part of the shape’s change compares with the unpredictable part.

The predictable part is the reversion. Its *direction* is fixed at any base date — which regions are stretched, and which way each must revert, is known today from the deviation of today’s shape from the norm. What grows with the horizon is not the direction but the *distance* travelled along it. Over a longer horizon the shape has moved further back toward the norm — roughly linearly at first, then with diminishing increments as it nears it.

The unpredictable part is news: the idiosyncratic repricing of individual regions as information arrives. It accumulates as a sum of many small, roughly independent daily shocks, and such a sum grows only like $\sqrt{h}$ — more slowly than reversion accumulates over the horizons studied.³

If reversion accumulates faster than news over the horizons studied, the signal-to-noise ratio of the realised cross-sectional pattern should rise across the tested range. At $h+10$ the reversion has barely moved the shape, so the realised change is mostly news and even a perfect forecaster would be ranking against a largely unrankable target; the IC is low for everyone. By $h+63$ the

²The exception is Day-Token, whose IC is flat across horizons.

³A sum of independent zero-mean shocks has variance equal to the sum of the per-step variances, so over h steps its variance grows like h and its typical size like $\sqrt{h}$; the standard random-walk scaling.

reversion has accumulated well clear of the news, a genuinely rankable pattern remains, and the IC is correspondingly high.

To confirm that this rise is a property of the surface rather than an artefact of the trained models, consider the simplest forecaster that exploits it: a parameter-free reversion rule.

Each surface is first de-levelled by subtracting its own cross-sectional mean across the 110 cells, giving $\tilde{S}_{c,t}$. The long-run shape $\bar{S}_c$ is the average of $\tilde{S}_{c,t}$ over the training set, a single fixed reference pattern carrying no test-period information. The rule's predicted change is the negative of today's deviation from that norm,

$$\hat{\Delta}_{c,t} = -(\tilde{S}_{c,t} - \bar{S}_c), \quad (4.1)$$

so cells stretched above their norm are predicted to fall and those below to rise, in proportion to how far each is stretched. The prediction has no fitted reversion speed and, decisively, no horizon argument: it is one vector, fixed at t , used unchanged whether the target is $h+10$ or $h+63$, and scored by the same rank IC as every model (Equation 3.31).

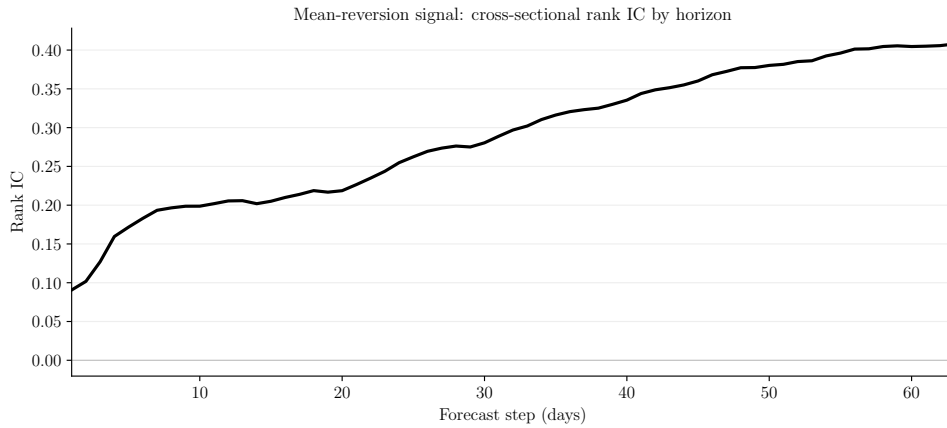


Figure 4.4: Cross-sectional rank IC of the parameter-free reversion rule over the test set, by forecast step.

Since the prediction is held fixed while only the target moves with h , any horizon dependence in its IC comes entirely from the target. Figure 4.4 confirms it: the rule's IC climbs monotonically with horizon from 0.09 at step 1, to around 0.41 by step 63.

This reframes the rule's role in what follows. Carried into the model comparison, it is the baseline that captures reversion alone; whatever a trained model scores above it is the cross-sectional structure that model has learned to anticipate *beyond* naive reversion.

Crucially, no model was trained to maximise IC; every one was trained to minimise MSE. Cross-sectional skill is thus not an objective any model pursued but a by-product of what each

learns to represent. Since the IC is level-unaware, the level shock that forced the crisis split of Section 4.2 does not distort it, and the comparison is made over the full test set.

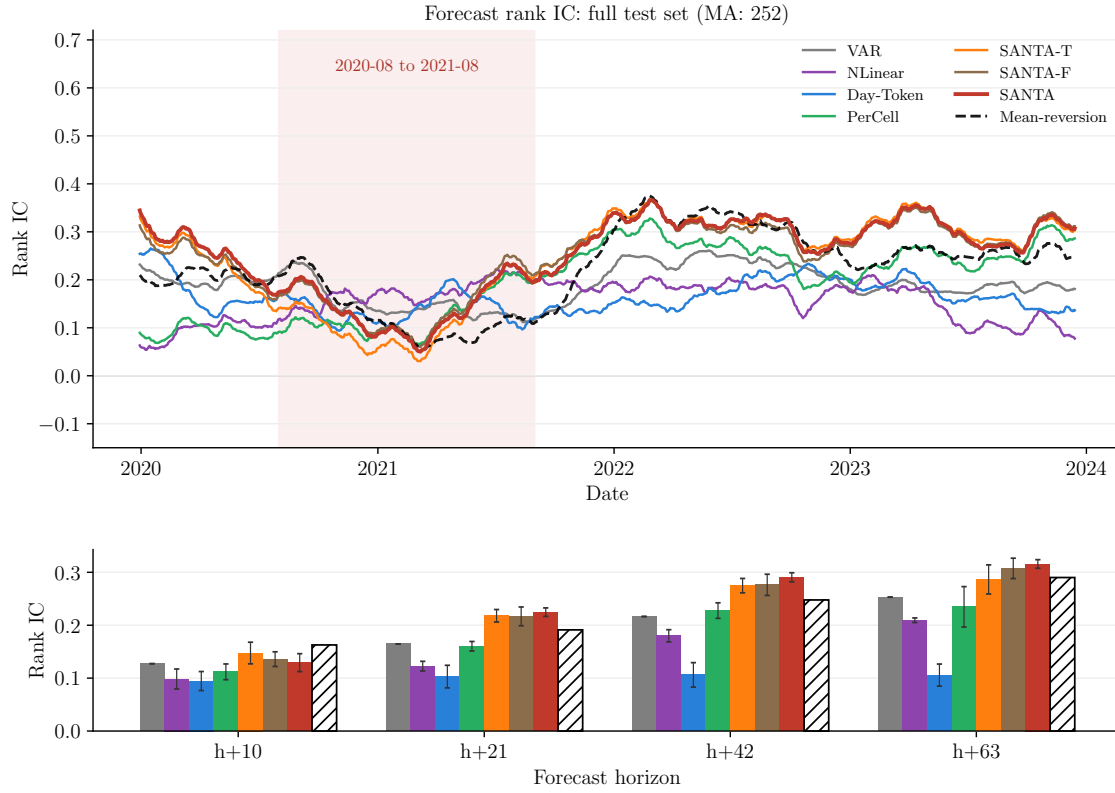


Figure 4.5: Cross-sectional rank IC over the full test set for every model and the parameter-free reversion rule, averaged across the ten seeds per model. Top: IC through time, as a 252-day moving average, with each forecast path averaged over all four horizons and assigned to its base date. Bottom: IC by horizon, with the seed standard deviation shown as whiskers.

The lower panel of Figure 4.5 reads the comparison by horizon. At $h+10$, where the realised shape change is mostly idiosyncratic shocks, no model recovers the cross-section better than the naive rule — its 0.16 edges the best model’s 0.15. At this shortest horizon the temporal-only SANTA-Temporal, which carries no spatial attention, posts the highest IC (0.147), ahead of the spatially-attentive SANTA (0.129) and SANTA-Flat (0.136). This is consistent with spatial attention fitting spurious cross-cell structure in the noise-dominated short-horizon target. At $h+21$ and $h+42$ all three SANTA variants outperform the rule, while by $h+63$ it is cleared by SANTA (0.316) and SANTA-Flat (0.307) but only matched by SANTA-Temporal (0.287 against the rule’s 0.290). At every horizon the reversion rule outperforms all non-SANTA models.

Cross-sectional shape skill beyond naive reversion is something only the SANTA architectures acquire; the remaining models do not learn to anticipate the surface’s shape any better than a rule that requires no learning at all. This points to the surface-aware embeddings, which only

the SANTA variants employ, as the source of the gain. Day-Token — the only transformer which flattens each day’s surface into a single vector, rather than maintaining a representation per cell — achieves an IC of 0.1 across all four horizons, the only model without a rising IC. The cross-cell structure the IC rewards is precisely what its day-token bottleneck discards at the input.

The upper panel traces IC through time, as a 252-day moving average over the full test set. For most of the period the SANTA family leads the field — a SANTA variant tops the line on roughly 60% of base dates — sitting above both the reversion rule and the remaining models. The exception is the window from roughly August 2020 to August 2021, in which the SANTA variants fall to the back of the field — and interestingly, the reversion rule falls with them. Over this window SANTA’s IC drops from its full-test average of 0.251 to 0.086, while the reversion rule’s falls from 0.220 to the same 0.086; the two reach their minimum on nearly the same day, with SANTA recovering faster afterward, regaining roughly 0.11 of IC within three months of the trough against the rule’s 0.02.

The reversion rule scores well only when the shape actually reverts toward the long-run norm estimated on the training data; its collapse over this window means that relationship itself broke down — the train-set shape-reversion ceased, for a time, to describe the market. SANTA’s skill tracks the same reversion structure, so when that structure failed, the SANTA variants fell with it. That it recovered faster than the rule suggests that what it captures is not reversion alone. A likely explanation is the unusual character of the post-COVID recovery over this window. Through 2020–21 the surface appears to have settled into a shape that departed from its historical norm and stayed there.

4.4 The Cost of Unconstrained Spatial Attention

The ladder’s final step replaces SANTA-Flat’s joint spatial attention with SANTA’s two axis-aligned blocks. The two models are matched on parameter budget and differ only in how they mix the surface across space: SANTA-Flat uses a single attention over all $M \cdot T = 110$ cells, free to place weight between any two of them, whereas SANTA splits the operation into two axis-wise attentions and can move weight along only one axis — moneyiness or maturity — at a time. Lifting the restriction does not shrink what the model can fit, so one might expect SANTA-Flat to match or beat SANTA. At the shorter horizons the two are indistinguishable on point-wise error. At the longer horizons the two separate — not reliably in mean error, but sharply in seed-to-seed variance.

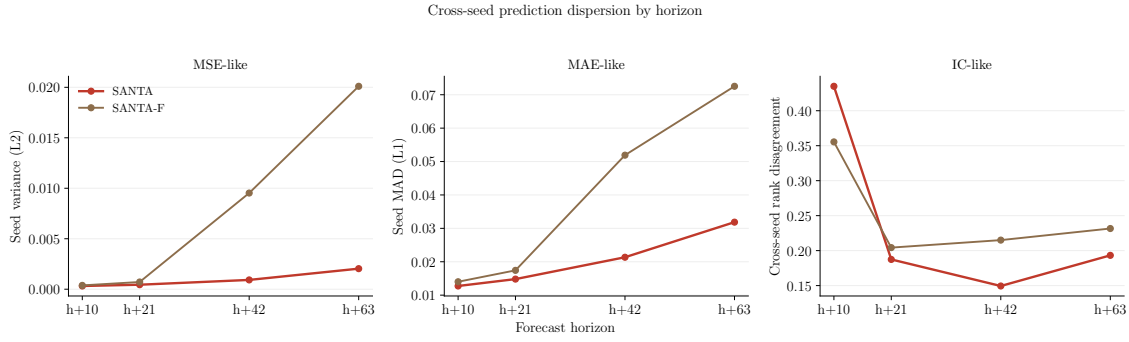


Figure 4.6: Cross-seed dispersion of SANTA and SANTA-Flat forecasts, by horizon, over the test set. Left: L_2 dispersion across the seeds; centre: L_1 dispersion; right: cross-seed rank disagreement.

Figure 4.6 reports the cross-seed dispersion of each model’s forecasts by horizon. At the shorter horizons SANTA and SANTA-Flat are indistinguishable on the point-wise (L_1 , L_2) measures; from $h+42$ onward SANTA-Flat’s seed-to-seed disagreement grows sharply while SANTA’s grows only gradually — roughly a ten-fold gap in squared dispersion by $h+63$, and more than twofold in absolute terms. Cross-seed rank disagreement, by contrast, stays within $1.4\times$ between the two models even at $h+42$ and $h+63$.

SANTA matches or beats SANTA-Flat across seeds despite its more restricted spatial mixing: the extra latitude buys no accuracy, only variance. Joint attention over all 110 cells leaves each seeded run more freedom to settle on its own solution, and independently seeded runs settle differently; SANTA’s two short axis-wise softmaxes (eleven- and ten-way) leave far less, and its runs stay close. The axial constraint is the better-matched prior — it supplies the surface’s two-axis structure directly rather than forcing each run to rediscover it, which is what the reproducibility gap measures.

4.5 Persistence as an Anchor

One of the design choices of this project was having every model in the ladder built to forecast a residual added to today’s surface, anchoring its prediction on persistence. The classical VAR carries no such anchor: it forecasts the surface’s coefficients directly and reconstructs the level from them. Figure 4.7 shows the consequence, tracing the per-step path error of each model relative to persistence along the $h+21$ horizon over the post-crisis period.

Every anchored model sits below persistence from the first forecast step and pulls steadily further ahead as the path lengthens. The VAR, by contrast, is worse than persistence for roughly

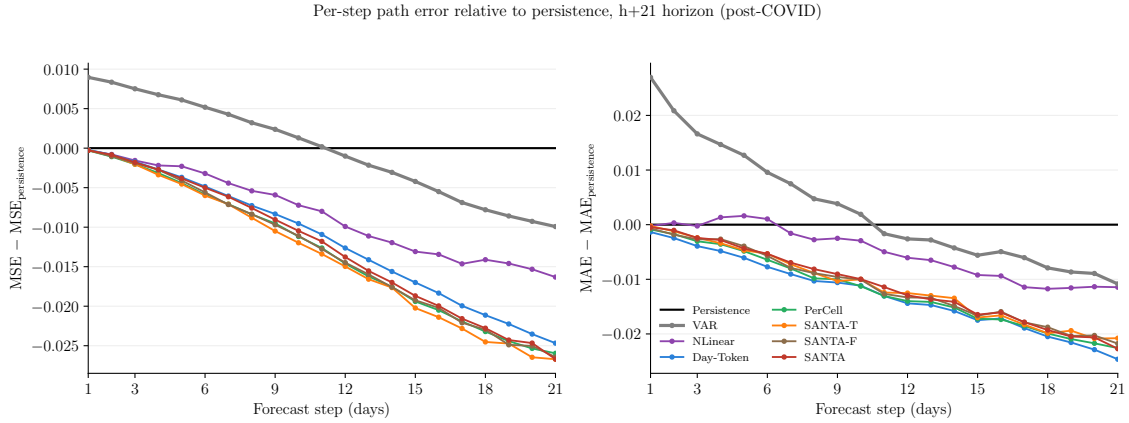


Figure 4.7: Per-step path error relative to persistence along the $h+21$ horizon (post-crisis period), averaged across the ten seeds per model. Left: MSE; right: MAE.

the first eleven steps. This early-path region is the benefit of anchoring: over the post-crisis period the VAR spends roughly two trading weeks earning back ground that a flat forecast holds for free.

4.6 Summary of Results

The two families of metric measure different things. Summed over the surface’s correlated cells, the point-wise errors (MSE, MAE) are dominated by how well a model forecasts the level and are only weakly sensitive to its shape; the rank IC, invariant to a common shift, scores the shape the point-wise metrics overlook. Across both, the field divides the same way — the surface-aware SANTA models on one side, and on the other the persistence and VAR benchmarks, the learned linear floor NLinear, and the geometry-agnostic Day-Token and Per-Cell transformers. This summary reads that division first, comparing SANTA against the rest of the field on each metric, before comparing within the SANTA family to weigh its variants against one another. On the point-wise errors the division is narrow; on the shape it is the principal result, and at the two longest horizon configurations SANTA records the best point estimate on every metric reported here.

On the point-wise errors the trained models are closely bunched, and it is the horizon that separates them. At $h+10$ the field is barely distinguishable from persistence — the trained architectures span 0.061–0.063 in post-crisis MSE against persistence’s 0.066 — holding today’s surface fixed being already near-optimal two trading weeks out. The gap widens along the path. Over the post-crisis period SANTA records the lowest MSE at both long horizons, 0.141 at $h+42$ and 0.168 at $h+63$, improving on persistence by roughly 16% at each and on the VAR by about 12%

at $h+63$. NLinear, though trained, ranks among the weakest models at these horizons, level with persistence and the VAR: as a static linear map it cannot adapt its response to the prevailing regime, and it trails all four per-cell transformers on roughly 72% of post-crisis base dates. The single pattern that reverses this ordering is the falling leg of a level collapse, the COVID recovery included, where the VAR’s reversion toward its fitted long-run level is rewarded and the anchored models — slower to commit to so abrupt a reversal — trail until the level settles.

On the shape the division is pronounced, and is best read against what the models were trained to do. Every model minimised MSE in standardised log-IV space, a loss the level dominates; none was trained to order the cross-section. Cross-sectional skill is therefore not an objective any model optimised directly, but a by-product of what its representation encodes, which makes the parameter-free reversion rule the natural reference: the ranking skill obtainable from the surface’s own tendency to revert toward a historical shape estimated on the training set. The reversion rule outperforms every non-SANTA model at every horizon — the VAR and Day-Token collapse the cross-section, while NLinear and the Per-Cell transformer represent each cell but give it no coordinate information. Only the SANTA variants clear the reference, and only from the one-month horizon outward. The split between the metrics is sharpest in Day-Token, whose post-crisis $h+63$ MSE is second only to SANTA’s yet whose IC sits flat, near 0.1 at every horizon — its day-token bottleneck discarding exactly the cross-cell structure the IC rewards. The source of the SANTA advantage is isolated by SANTA-Temporal, identical to the Per-Cell transformer but for the coordinate embeddings: the two stand at 0.22 against 0.16 at $h+21$ and 0.29 against 0.23 at $h+63$, the embeddings alone carrying the model from below the reference, where Per-Cell remains, to meet or exceed it.

Turning within the SANTA family, the gains are smaller, and the comparison turns on the appropriate prior rather than accuracy alone. SANTA-Temporal, which moves no information between cells, attains most of what the full model does — within roughly 2% of SANTA’s long-horizon MSE and about 0.03 of its IC, and at $h+10$ the stronger shape forecaster, at 0.147 against 0.129. Unconstrained spatial attention adds variance rather than accuracy: SANTA-Flat’s seed-to-seed dispersion grows by roughly an order of magnitude at the long horizons, its $h+63$ MSE standard deviation reaching ± 0.036 against SANTA’s ± 0.002 , and its mean error drifting to 0.190 where SANTA holds 0.168. Factoring the same attention along the moneyiness and maturity axes restores the stability, and SANTA betters the unconstrained ablation on all three metrics at $h+42$ and $h+63$. The axial constraint is therefore the better-matched prior.

These point estimates favour SANTA throughout, but the statistical support is uneven (Ap-

pendix C.5). What holds firmly is the contrast at the centre of this summary: SANTA-Temporal’s IC advantage over the otherwise-identical Per-Cell transformer — the cleanest test of the coordinate embeddings — is significant at every horizon (at the 10% level or better, and at 5% at three of four). The finer distinctions are not, for two reasons that compound. The trained models produce point-wise errors of similar magnitude, so the differences under test, within the SANTA family and between it and the rest alike, are small beside their variability; and the overlapping forecast windows shrink the effective sample from around one hundred at $h+10$ to roughly fifteen at $h+63$ — the horizon at which SANTA’s point-estimate lead is largest and the test’s power is lowest. Where the effect is large, as on the IC gap to the models without coordinate embeddings, it survives; where it is a narrow point-wise margin between closely matched models, the sample is too thin to confirm it.

5

Evaluation

Contents

5.1 Objectives Revisited	61
5.2 Key Empirical Findings	62
5.3 Limitations	62

5.1 Objectives Revisited

This project set out to do something the literature had largely left open: to forecast the full implied volatility surface, multiple steps ahead, from its own history alone, and to ask whether respecting the surface’s structure helps. Direct multi-horizon forecasting of the whole surface is a thin strand of an otherwise large field, and no attention-based model forecasting the surface multi-step from its own past was found in the literature review of Chapter 2. The work therefore had to build its own ground before it could stand on it.

Four deliverables were set, and each was met. A clean surface dataset was constructed. A surface-aware tensor-attention architecture, SANTA, was designed and built. And it was placed in a controlled ladder of ablations, isolating one structural ingredient at a time at matched capacity and shared hyperparameters, attributing any gaps in performance to architectural differences. Every model on that ladder, together with the classical benchmarks, was tested under one protocol.

5.2 Key Empirical Findings

The central finding is that SANTA holds both ends of the problem at once, where no other model does. On point-wise error it is competitive from the first forecast step and strongest at the long horizons, though its lead over the trained field is thin. On cross-sectional IC the separation is real: the SANTA variants are the only learned models to clear the field, SANTA reaching 0.32 at the longest horizon against 0.25 for the best non-SANTA model — a gap several times the one separating the whole trained field on MSE and MAE. Models that score low point-wise errors cannot order the cross-section as well; SANTA does both. That balance, rather than any single decisive margin, is what the study establishes.

These outcomes turn on three design decisions, which the ladder was built to isolate and which earn their place unequally. Anchoring every forecast on today’s surface builds the surface’s near-martingale short-horizon behaviour into the prediction, so every anchored model improves on persistence from the first step on average, where the unanchored VAR spends its first two trading weeks with larger errors. The shape advantage traces to the coordinate embeddings: SANTA-Temporal differs from the Per-Cell transformer by nothing else, and the IC gain over that otherwise-identical model holds at every horizon. The factored spatial attention earns the least: SANTA-Temporal, with no spatial attention at all, finishes within about 2% of the full model on point-wise error and 0.03 on IC. What factoring buys is not accuracy but stability — constraining attention to the two surface axes recovers roughly an order of magnitude in seed reproducibility at the longer horizons over freely joint spatial attention.

The advantage is not unconditional: through 2020–21 the surface’s shape ceased to revert toward its training-set norm, and SANTA’s skill fell with it to the back of the field.

5.3 Limitations

The bounds on these conclusions are of two kinds: limits on what the evidence can carry, and scope drawn deliberately.

The evidence is narrow. One underlying was studied — the SPX — and only over a four-year test window. Each regime-level conclusion is thinner still — any regime-dependent observations support a mere description, not generalisation. The same thinness reaches the statistical significance tests: the overlapping forecast windows shrink the effective sample size considerably at the

longer horizons, precisely where SANTA’s point-estimate lead is widest, so the margins that look largest are the ones the significance tests are least able to confirm.

No economic value analysis was conducted. The forecasts were never carried into a trading or hedging exercise, and the survival of the forecasting accuracy edge presented under transaction costs was not tested. All evaluation metrics are reported in standardised log-IV, the right space for comparing the models, but one in which the translation of the reported gains in volatility points, or in money, is left open.

Finally, the controlled comparison was built around the decision not to hyperparameter tune any of the models. As such, the reported performance can generally be considered a floor on what the architecture could reach. It is untested whether the ladder’s ordering would survive a search over depth, width, lookback, or schedule. The instability of the unconstrained spatial attention, in particular, may belong to this fixed budget rather than to a hyperparameter space a thorough search could reveal. The findings concern which priors matter at matched scale, and should be read as no more.

The remaining boundaries were chosen rather than imposed. The forecast is unconditional, drawing on the surface’s own history and nothing else — the index return, the VIX, and the leverage effect (Section 2.2.2) are all excluded — so the results stand as a lower bound on what a conditioning model could reach. And the object forecast is the market-made surface, with no arbitrage conditions imposed on inputs or outputs, so the predictions serve analysis rather than direct pricing. Both choices define what the project is: a controlled first account of which structural priors earn their keep in a setting the literature had left open.

Conclusions and future directions

Conclusions

This project asked whether the implied volatility surface can be forecast multiple steps ahead from its own history, and whether a model that respects its structure forecasts it better than one that does not. The answer is a qualified yes, and SANTA is what carries it.

SANTA was not reached in one step but assembled — each rung of the ladder adding a single structural idea, and each justified by what its absence costs. Per-cell representation, coordinate awareness, and attention factored along the surface's own axes are not competing designs but the parts of one, and the ablations stand as the evidence that each earns its place. What results is the model that, at the long horizons, posts the best score on every metric the study reports. That breadth, more than any single margin, is the case for building the surface's structure into the model rather than leaving it to be recovered from data.

Yet the surface is only partially forecastable from its own past: most of its movement, driven by external innovations, stays beyond reach.

What remains is a careful map of a setting the literature had largely left open: which structural priors earn their place in forecasting the surface, where their advantage lives, and where it gives out. The map is drawn from one underlying over one window, and thus reads as a floor rather than a verdict — a starting point, offered to whoever forecasts this surface next.

Future Directions

The clearest extension is economic. Performance here was measured in standardised log-IV, not in money: the forecasts were never used to hedge or trade, so it is unknown whether SANTA's edge is large enough to cover transaction costs. This matters because the edge is small, and small statistical gains often vanish once frictions are charged against them. A study that runs the forecasts through a hedging or trading rule, and reports the result net of costs, would establish

whether the accuracy shown here has any economic value at all.

Three further extensions follow from choices made in the setup. Repeating the comparison on other index and individual-equity surfaces would show whether the rankings hold beyond SPX alone. Conditioning on the exogenous drivers could lift forecasts above the unconditional bound this work reports. And a search over the deep models' hyperparameters, fixed throughout for controlled comparison, could shift both the absolute numbers and the ordering of the ladder.



Data Implementation Details

This appendix collects the replication-grade details of the data pipeline.

A.1 Data pipeline

The pipeline consumes two OptionMetrics files: the option-price file recording every traded SPX contract on each day, and the forward-price file giving the implied forward price of the S&P 500 at each future expiration. The fields used from each are listed in Table A.1.

Table A.1: OptionMetrics fields used in this work.

File	Field	Role
Option prices	date	Trade date
	exdate	Contract expiration date
	cp_flag	Call/put indicator
	strike_price	Strike (stored $\times 1000$; rescaled on read)
	impl_volatility	Vendor-supplied implied volatility
	vega	Surface-fit weighting; quality filter
	volume	Quality filter
Forward prices	date	Trade date (join key)
	exdate	Expiration date (join key)
	ForwardPrice	Implied forward price for the expiry
	AMSettlement	AM/PM settlement flag; PM retained where both exist

The two files are joined on (date, expiration). Where both an AM- and a PM-settled forward exist for the same expiry, the PM value is retained, since SPX index options are PM-settled by convention and the PM forward is the standard reference. On days where only the AM forward is

recorded, it is used in place of the PM. The resulting sub-day timing discrepancy at AM expiries is a known minor source of carry error.

OptionMetrics records mid-quotes at 15:59 ET against the official composite close of the index. This is the snapshot used for all surface construction in this work.

A.2 Filter thresholds

The four filter families of Section 3.1.2 are applied with the thresholds in Table A.2.

Table A.2: Filter thresholds applied during surface construction. Moneyness m is log-forward-moneyness, $m = \log(K/F)$. Maturity τ is in years (calendar-day count divided by 365).

Family	Filter	Threshold
Scope	Moneyness window	$m \in [-0.25, +0.25]$
	Maturity window	$\tau \in (0, 1.75]$ years
Validity	IV bracket	$\sigma \in (0.01, 3.0)$
	IV finite	required
	Vega finite	required
Liquidity	Vega floor (core)	$\nu \geq 0.5$
	Vega floor (wings)	$\nu \geq 0.1$
	Volume floor (core)	≥ 1 contract
	Volume floor (wings)	waived
	Wings defined as	$ m > 0.07$
OTM convention	Calls kept for	$m \geq 0$
	Puts kept for	$m \leq 0$
	ATM band	$ m < 0.01$ admits both

The scope thresholds are set generous around the target grid of Section 3.1.4, which uses $m \in [-0.10, +0.10]$ and $\tau \in [30, 365]$ days, so the smoother has neighbouring quotes well outside the grid's edges on every side.

A.3 Smoothing configuration

The Nadaraya–Watson estimator of Section 3.1.3 is applied with the configuration in Table A.3. The bandwidths h_τ and h_m are kernel variances in the Gaussian kernel $g(x, y) \propto \exp(-\frac{1}{2}(x^2/h_\tau + y^2/h_m))$, so the corresponding standard deviations are $\sqrt{h_\tau}$ in $\log \tau$ units and $\sqrt{h_m}$ in m units.

Cells failing either diagnostic guard are filled by their immediate grid neighbours via forward- and back-fill along both axes.

Table A.3: Smoothing configuration: bandwidth values and diagnostic thresholds. Bandwidths are kernel variances.

Parameter	Value	Role
h_τ	0.16	Maturity bandwidth (constant across surface)
h_m at $\tau = 30$ days	2.0×10^{-4}	Moneyness bandwidth at short end
h_m at $\tau = 365$ days	6.0×10^{-4}	Moneyness bandwidth at long end
$h_m(\tau)$ interpolation	linear in $\log \tau$	—
n_{eff} floor	2.0	Kish effective sample size
Max single-quote share	0.85	Concentration guard

B

Model Architectural Details

The five deep architectures share the following specifications: two encoder layers, four attention heads, an FFN with hidden dimension equal to the embedding dimension, and dropout of 0.1. The lookback length is $L = 63$ surfaces and the surface grid is 11 moneyness points by 10 maturities, giving 110 cells per surface. NLinear shares the lookback and grid but carries none of this structure: it is a single linear map, included in the tables below for completeness. Per-model architectural choices are summarised in Table B.1. Parameter counts at each horizon are given in Table B.2.

Table B.1: Per-model embedding and head dimensions. The Day-Token model’s embedding dimension is shown for $h = 10, 21, 42, 63$ in order; the other four deep models hold d fixed across horizons. NLinear has no embedding or attention structure.

Model	Embedding dim d	Head hidden dim
NLinear	—	—
Day-Token	24 / 16 / 8 / 8	—
Per-Cell	56	24
SANTA-Temporal	56	24
SANTA-Flat	40	24
SANTA	32	24

The Day-Token is the only deep model whose embedding dimension varies with horizon: its single output head is a linear map of dimension $d \times 110h$, which grows linearly with h , so d is reduced as the horizon lengthens to keep the total parameter count in a comparable range. The other four deep architectures use a per-cell output head whose parameter count grows only slowly with h , so d is fixed across horizons. At the longest horizon $h = 63$, even with d floored at 8, the Day-Token’s head dominates and its parameter count rises to 66,418. NLinear’s count is simply that of its weight matrix, $h \times (L + 1) = 64h$; it grows linearly with the horizon but, being shared across all

110 cells, is independent of the grid and negligible against the deep models.

Table B.2: Parameter counts per model and horizon.

Model	$h = 10$	$h = 21$	$h = 42$	$h = 63$
NLinear	640	1,344	2,688	4,032
Day-Token	44,252	48,902	45,628	66,418
Per-Cell	43,722	43,997	44,522	45,047
SANTA-Temporal	50,330	50,605	51,130	51,655
SANTA-Flat	46,762	47,037	47,562	48,087
SANTA	43,490	43,765	44,290	44,815

C

Supplementary Results

C.1 Sample Forecasts

The volume of forecast surfaces is too large to illustrate in the main text, so an array of examples is collected here, leaving Chapter 4 to focus on trends and aggregate performance. For each horizon configuration $h \in \{10, 21, 42, 63\}$, one forecast path is selected at random from the test set and its final-step forecast is plotted against the realised surface at the target date. All forecasts are produced by trained models with seed 0. The surfaces are shown in de-standardised raw-IV space, and the accompanying error metrics are also computed in this space.

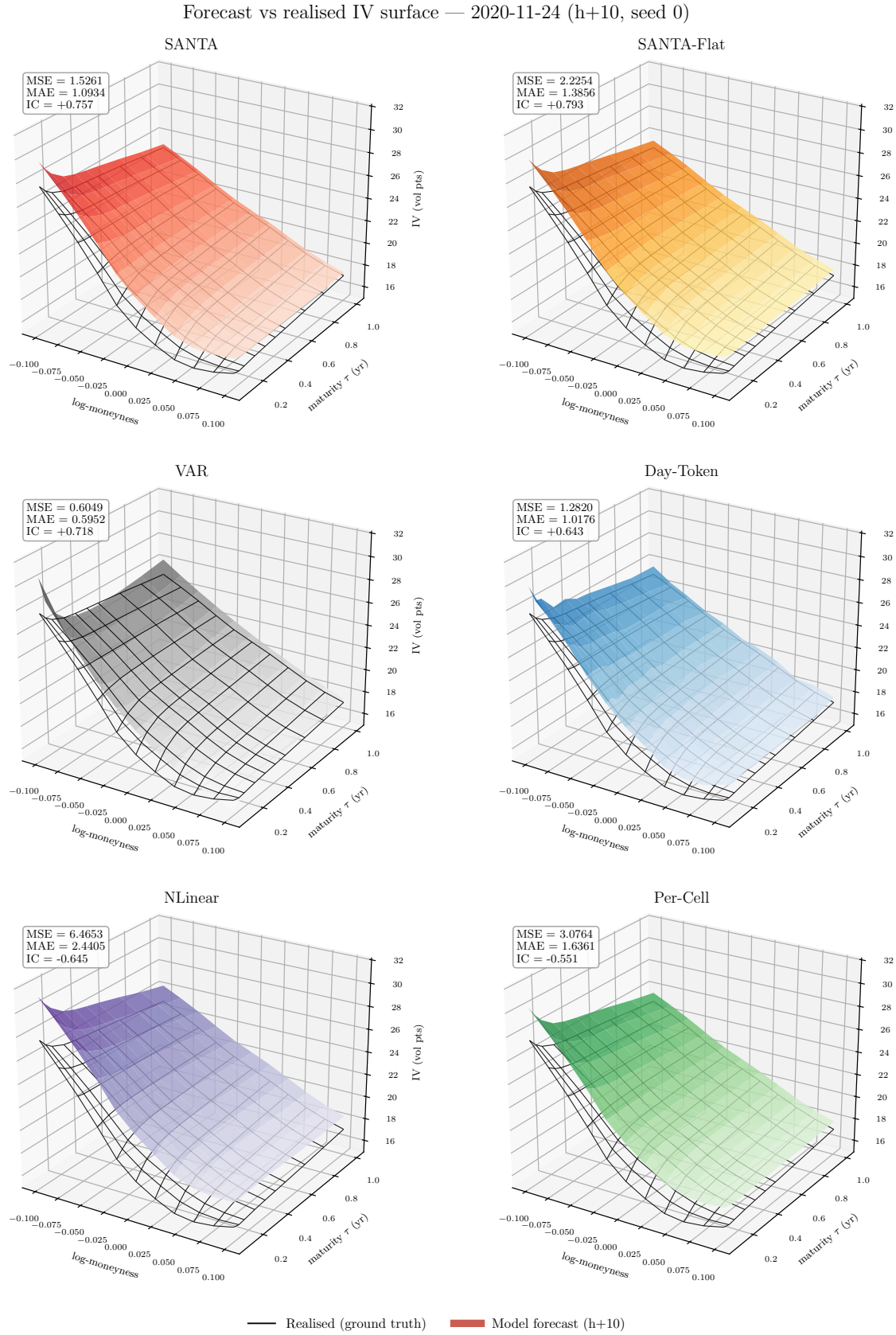


Figure C.1: Forecast versus realised surface at the final step of a randomly selected $h + 10$ path; target date 24 November 2020.

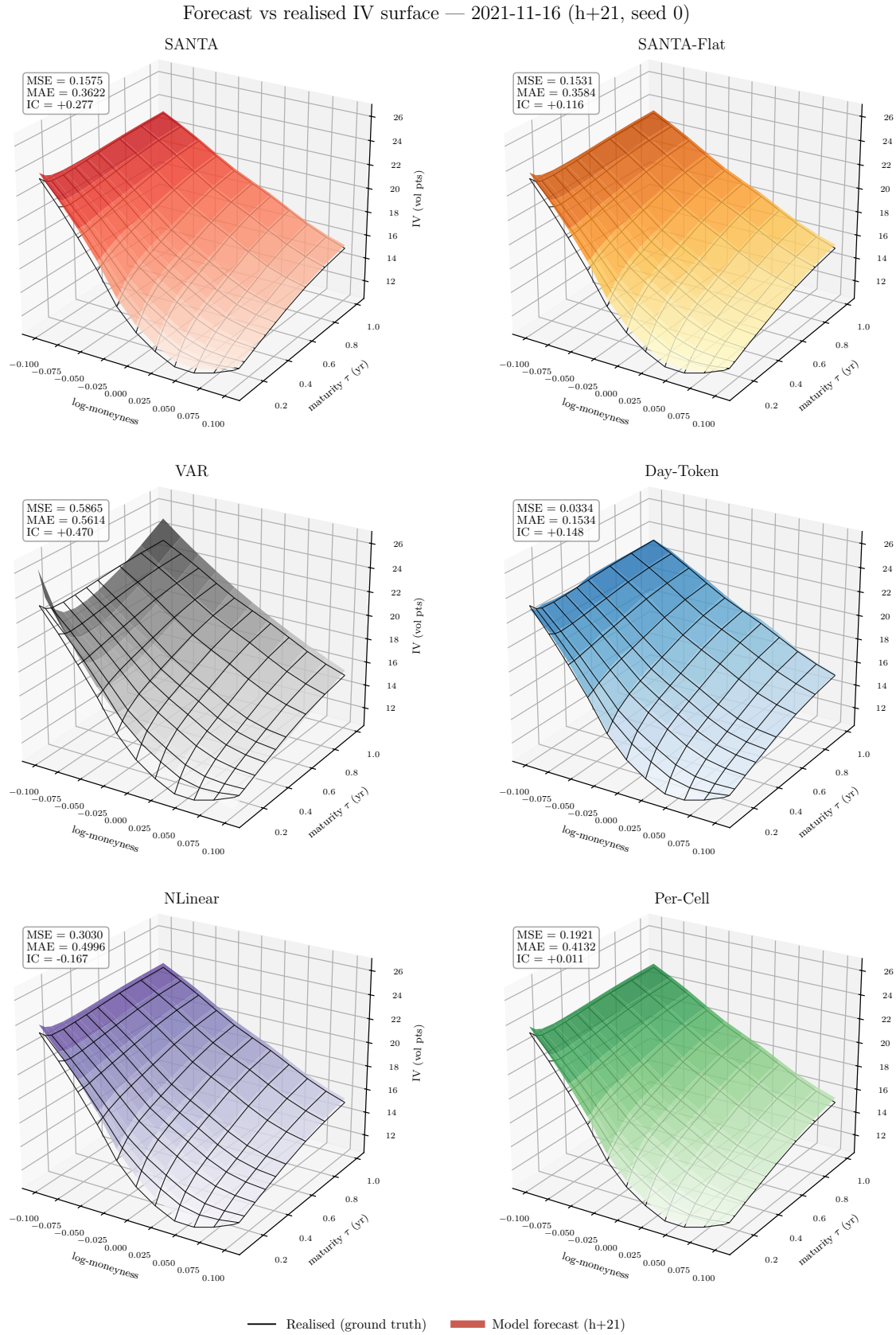


Figure C.2: Forecast versus realised surface at the final step of a randomly selected $h + 21$ path; target date 16 November 2021.

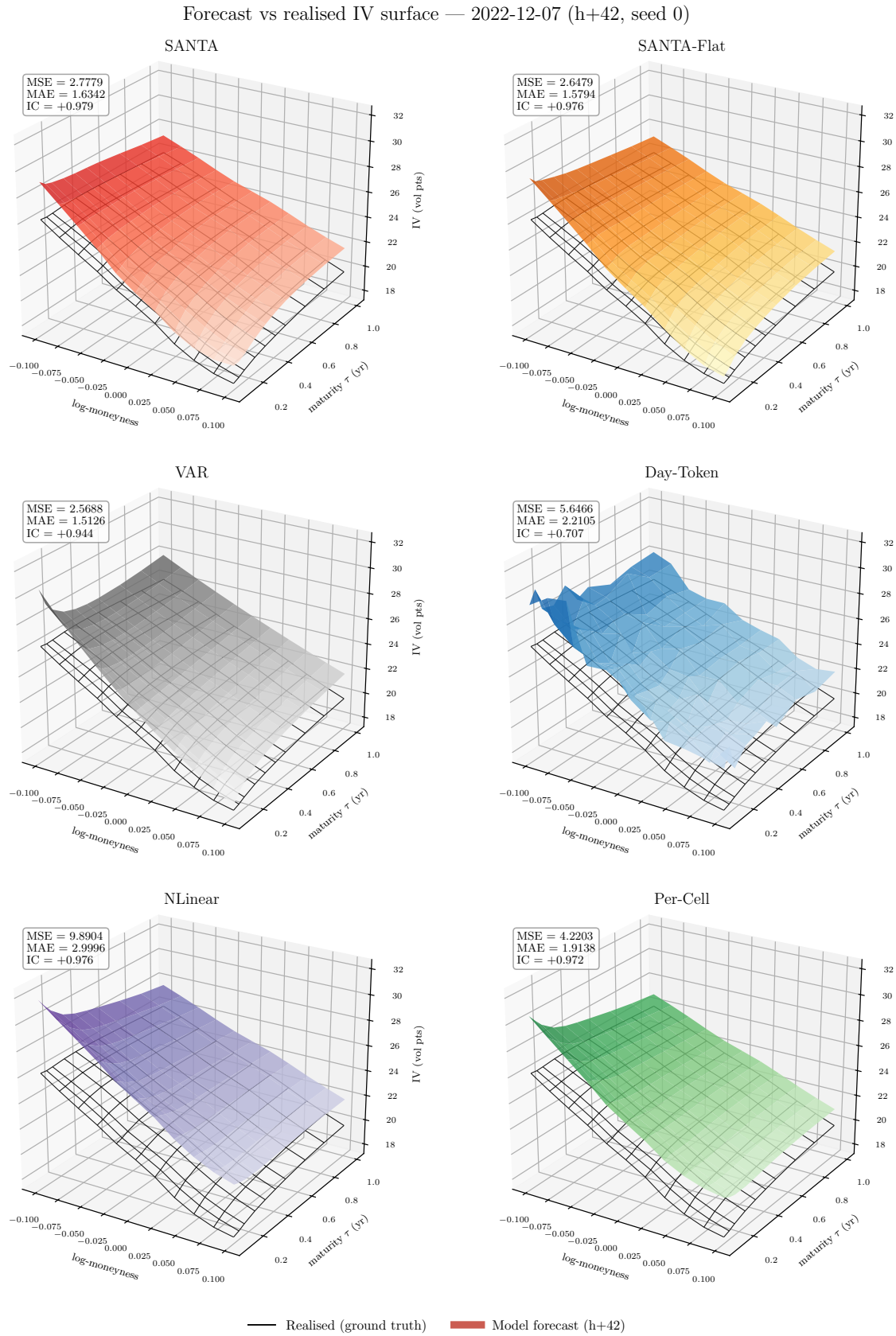


Figure C.3: Forecast versus realised surface at the final step of a randomly selected $h + 42$ path; target date 7 December 2022.

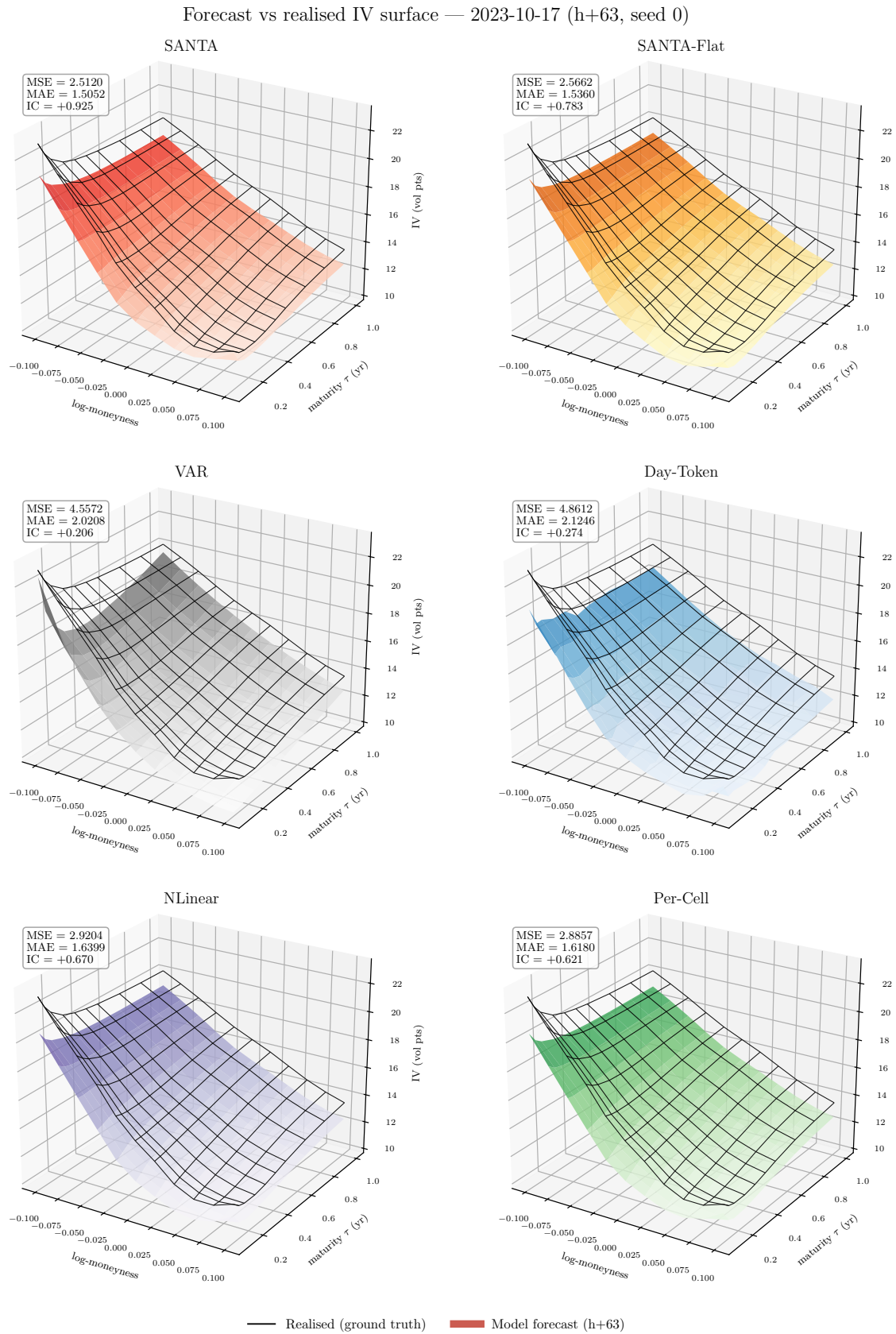


Figure C.4: Forecast versus realised surface at the final step of a randomly selected $h + 63$ path; target date 17 October 2023.

C.2 Crisis and Post-crisis Period Aggregate Performance Tables

This section reproduces the aggregate accuracy table of Chapter 4, restricted to each side of the split introduced in Section 4.2: the COVID crisis window (forecast base dates 31 December 2019 to 30 April 2020) and the post-crisis remainder (May 2020 onwards). Each table reports the same models, metrics, and seed treatment as the full-test version; only the set of forecast windows differs.

Horizon	Model	MSE	MAE	IC
$h+10$	Persistence	0.57205	0.49506	—
	VAR	0.63519	0.49722	0.2587
	NLinear	0.63289 ± 0.0216	0.49596 ± 0.0149	0.1054 ± 0.042
	Day-Token	0.61487 ± 0.0055	0.49613 ± 0.0041	0.1175 ± 0.027
	Per-Cell	0.60346 ± 0.0081	0.50022 ± 0.0050	-0.0091 ± 0.035
	SANTA-T	0.59774 ± 0.0154	0.49873 ± 0.0103	<u>0.1537 ± 0.103</u>
	SANTA-F	0.58432 ± 0.0070	<u>0.48927 ± 0.0035</u>	0.0720 ± 0.133
	SANTA	<u>0.57953 ± 0.0070</u>	<u>0.48816 ± 0.0038</u>	0.0815 ± 0.052
$h+21$	Persistence	1.41394	0.77147	—
	VAR	1.52364	0.74471	0.2790
	NLinear	1.45395 ± 0.0202	0.74399 ± 0.0121	-0.0022 ± 0.044
	Day-Token	1.44932 ± 0.0201	0.74987 ± 0.0069	0.1452 ± 0.018
	Per-Cell	1.40200 ± 0.0119	0.74327 ± 0.0061	-0.0587 ± 0.048
	SANTA-T	1.39602 ± 0.0371	0.74102 ± 0.0222	0.3878 ± 0.102
	SANTA-F	<u>1.39093 ± 0.0184</u>	<u>0.73747 ± 0.0084</u>	0.3340 ± 0.120
	SANTA	<u>1.38022 ± 0.0126</u>	<u>0.73365 ± 0.0057</u>	<u>0.3807 ± 0.021</u>
$h+42$	Persistence	2.92332	1.23002	—
	VAR	2.99228	1.14830	0.2987
	NLinear	2.79960 ± 0.0378	1.13972 ± 0.0226	0.0433 ± 0.056
	Day-Token	2.86993 ± 0.0231	1.16687 ± 0.0100	0.1628 ± 0.033
	Per-Cell	2.73979 ± 0.0297	1.14025 ± 0.0074	0.0575 ± 0.080
	SANTA-T	<u>2.72616 ± 0.0596</u>	<u>1.13467 ± 0.0163</u>	0.4839 ± 0.038
	SANTA-F	2.83230 ± 0.1416	1.17895 ± 0.0523	0.4300 ± 0.053
	SANTA	<u>2.68792 ± 0.0297</u>	<u>1.12831 ± 0.0099</u>	<u>0.4821 ± 0.054</u>
$h+63$	Persistence	3.59735	1.45290	—
	VAR	3.54198	1.31979	0.3578
	NLinear	3.30643 ± 0.0355	1.29659 ± 0.0240	0.0994 ± 0.059
	Day-Token	3.45039 ± 0.0409	1.35475 ± 0.0259	0.1916 ± 0.031
	Per-Cell	3.22360 ± 0.0453	1.30209 ± 0.0114	0.1868 ± 0.079
	SANTA-T	<u>3.20419 ± 0.0461</u>	<u>1.28955 ± 0.0168</u>	<u>0.5177 ± 0.036</u>
	SANTA-F	3.23018 ± 0.1004	1.30458 ± 0.0350	0.5039 ± 0.067
	SANTA	<u>3.17829 ± 0.0736</u>	<u>1.29655 ± 0.0156</u>	<u>0.5457 ± 0.065</u>

Table C.1: Test-set accuracy across horizons (standardised log-IV), **COVID crisis period** (forecast base date 31 Dec 2019 – 30 Apr 2020). Trained models reported as mean $\pm$ sd over 10 seeds. Best per horizon and metric in **bold**, second underlined.

Horizon	Model	MSE	MAE	IC
$h+10$	Persistence	0.06566	0.18521	—
	VAR	0.07116	0.19731	0.1151
	NLinear	0.06284 ± 0.0007	0.18573 ± 0.0011	0.0977 ± 0.018
	Day-Token	0.06188 ± 0.0007	0.17995 ± 0.0008	0.0924 ± 0.019
	Per-Cell	<u>0.06150 ± 0.0013</u>	<u>0.18088 ± 0.0022</u>	0.1232 ± 0.015
	SANTA-T	0.06117 ± 0.0012	0.18088 ± 0.0013	0.1468 ± 0.016
	SANTA-F	0.06226 ± 0.0007	0.18204 ± 0.0011	<u>0.1419 ± 0.010</u>
	SANTA	0.06237 ± 0.0006	0.18110 ± 0.0007	0.1337 ± 0.019
$h+21$	Persistence	0.11166	0.24921	—
	VAR	0.11146	0.25183	0.1539
	NLinear	0.10395 ± 0.0010	0.24502 ± 0.0010	0.1344 ± 0.010
	Day-Token	0.10162 ± 0.0009	0.23845 ± 0.0014	0.0989 ± 0.023
	Per-Cell	0.09942 ± 0.0019	0.23801 ± 0.0025	0.1809 ± 0.010
	SANTA-T	0.09896 ± 0.0020	0.23872 ± 0.0025	0.2018 ± 0.008
	SANTA-F	<u>0.09916 ± 0.0015</u>	<u>0.23831 ± 0.0018</u>	<u>0.2057 ± 0.009</u>
	SANTA	0.09942 ± 0.0008	0.23835 ± 0.0011	0.2100 ± 0.008
$h+42$	Persistence	0.16828	0.31537	—
	VAR	0.16021	0.30855	0.2085
	NLinear	0.15584 ± 0.0024	0.30676 ± 0.0016	0.1932 ± 0.014
	Day-Token	0.14889 ± 0.0017	0.29566 ± 0.0016	0.1007 ± 0.023
	Per-Cell	0.14320 ± 0.0021	<u>0.29276 ± 0.0019</u>	0.2441 ± 0.013
	SANTA-T	<u>0.14218 ± 0.0047</u>	0.29282 ± 0.0049	0.2546 ± 0.014
	SANTA-F	0.14842 ± 0.0119	0.29634 ± 0.0093	<u>0.2615 ± 0.021</u>
	SANTA	0.14074 ± 0.0008	0.29063 ± 0.0009	0.2720 ± 0.010
$h+63$	Persistence	0.20062	0.34791	—
	VAR	0.19014	0.33704	0.2427
	NLinear	0.19078 ± 0.0020	0.34335 ± 0.0017	0.2204 ± 0.007
	Day-Token	0.17810 ± 0.0018	0.32548 ± 0.0021	0.0973 ± 0.021
	Per-Cell	0.18303 ± 0.0193	0.33139 ± 0.0147	0.2395 ± 0.039
	SANTA-T	<u>0.17392 ± 0.0070</u>	<u>0.32527 ± 0.0060</u>	0.2637 ± 0.029
	SANTA-F	0.18954 ± 0.0356	0.33233 ± 0.0223	<u>0.2880 ± 0.015</u>
	SANTA	0.16806 ± 0.0020	0.32008 ± 0.0024	0.2930 ± 0.013

Table C.2: Test-set accuracy across horizons (standardised log-IV), **post-crisis period** (forecast base date from 1 May 2020 onwards). Trained models reported as mean $\pm$ sd over 10 seeds. Best per horizon and metric in **bold**, second underlined.

C.3 Post-crisis MAE

This appendix reproduces the post-crisis point-wise error analysis figure of Section 4.2.1 under MAE. The qualitative picture is unchanged: error grows with the horizon, the trained architectures bunch at the short horizons and separate into a ranking from $h+42$ onward, SANTA-Flat shows the widest seed standard deviation at the longest horizons while the factored SANTA stays tight, and the time path shows the same level-timed reversal in which VAR and Day-Token lead on sharp falls in the surface level and trail when it rises or stays high.

While the rankings produced by the bar chart associated with MSE are maintained, the differences are much smaller.

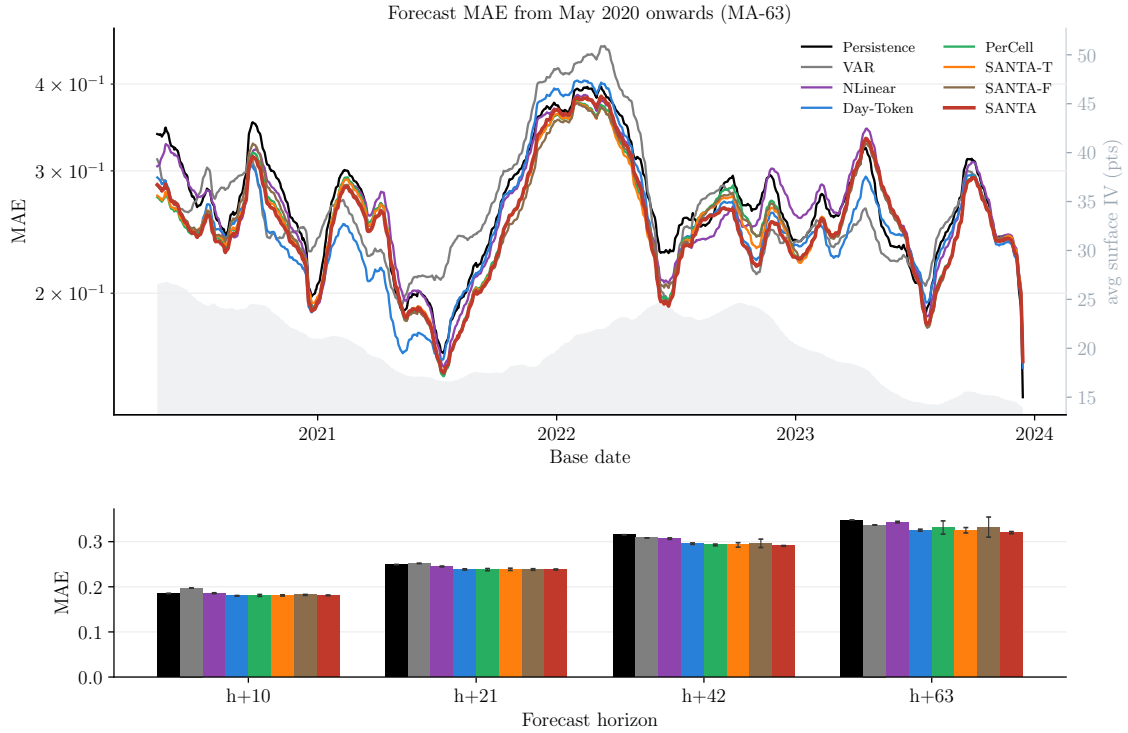


Figure C.5: Forecast MAE over the post-crisis period (May 2020 onward), averaged across the ten seeds per model. Top: 63-day moving average through time, with each forecast path averaged over all four horizons and assigned to its base date. Bottom: MAE by horizon, with the seed standard deviation shown as whiskers.

C.4 Crisis-period MAE

This appendix reproduces the crisis-period point-wise error analysis figure of Section 4.2.2 under MAE. Again, the qualitative picture is unchanged: errors are an order of magnitude larger than in the post-crisis period, the SANTA family carries the lowest error through the run-up to the peak and VAR leads the recovery as the level falls.

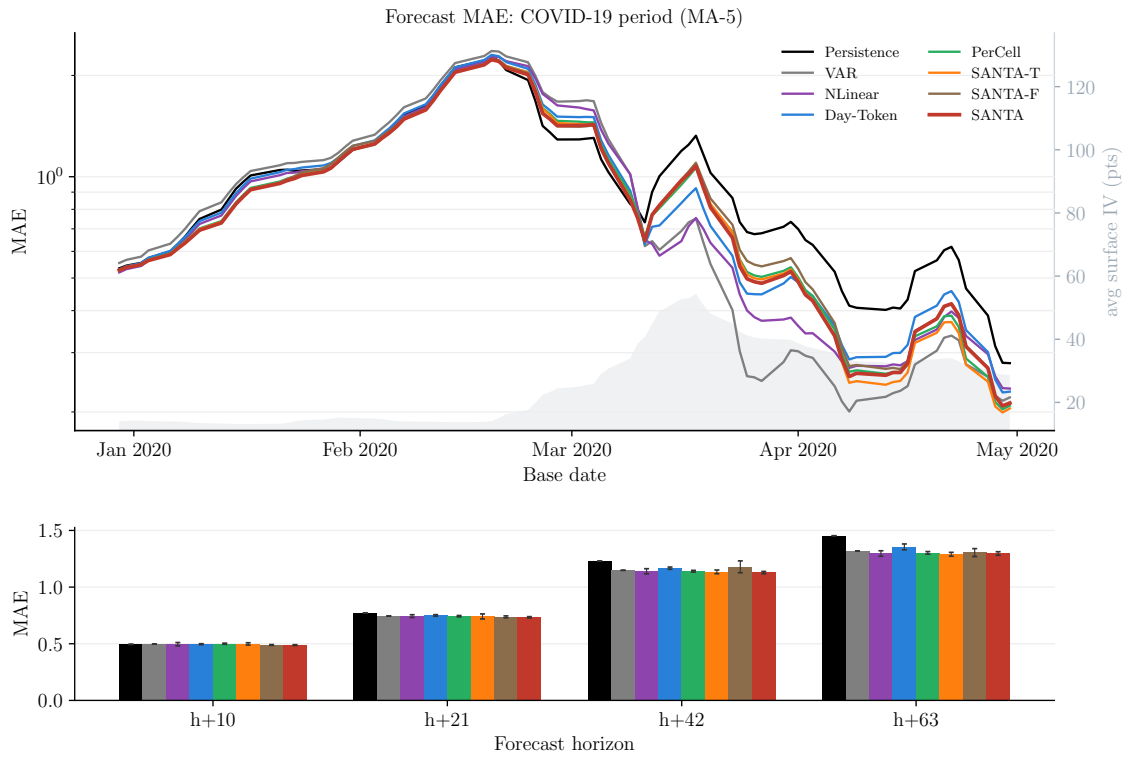


Figure C.6: Forecast MAE over the COVID-19 crisis period, averaged across the ten seeds per model. Top: 5-day moving average through time, with each forecast path averaged over all four horizons and assigned to its base date, and the average level of the SPX surface overlaid. Bottom: MAE by horizon, with the seed standard deviation shown as whiskers.

C.5 Statistical Significance Supplementary Results

The Diebold–Mariano test [34] assesses whether two models have equal expected predictive accuracy under a chosen loss. For models a and b , the per-window loss differential $d_t = L_t^{(a)} - L_t^{(b)}$ is formed, and the null of equal accuracy, $\mathbb{E}[d_t] = 0$, is tested through the statistic

$$\text{DM} = \frac{\bar{d}}{\sqrt{\widehat{\text{Var}}(\bar{d})}}, \quad (\text{C.1})$$

the mean differential standardised by its long-run standard error, with a positive value favouring model a .

Two adjustments are made for the structure of the data. The forecast windows overlap — a horizon- h path shares $h-1$ steps with each neighbour — so the differentials are serially correlated; the long-run variance is therefore estimated with the Newey–West heteroskedasticity- and autocorrelation-consistent estimator [35], with truncation lag $h-1$ to match the overlap. This means the effective number of independent observations is roughly N/h , falling from about 100 at $h+10$ to 15 at $h+63$, where the asymptotic test is over-sized. The Harvey–Leybourne–Newbold correction [36] is applied to address this. All reported significance uses this corrected form, at the 10%, 5% and 1% levels. Persistence is excluded.

	VAR	NLinear	Day-Token	Per-Cell	SANTA-T	SANTA-F
$h+10$ (<i>eff. N</i> ≈ 100)						
NLinear	+3.15***					
Day-Token	+5.10***	+0.87				
Per-Cell	+3.37***	+0.91	−0.39			
SANTA-T	+3.31***	+1.04	+0.79	+1.61		
SANTA-F	+3.10***	+0.91	+0.55	+0.55	−0.39	
SANTA	+2.97***	+0.87	+0.59	+0.59	−0.04	+0.48
$h+21$ (<i>eff. N</i> ≈ 47)						
NLinear	+1.60					
Day-Token	+2.05**	+0.80				
Per-Cell	+1.68*	+1.21	+0.96			
SANTA-T	+1.70*	+1.29	+1.04	+1.28		
SANTA-F	+1.70*	+1.27	+1.04	+0.82	+0.46	
SANTA	+1.68*	+1.23	+1.05	+0.82	−0.01	+0.54
$h+42$ (<i>eff. N</i> ≈ 23)						
NLinear	+1.01					
Day-Token	+1.42	+0.17				
Per-Cell	+1.38	+1.70*	+1.20			
SANTA-T	+1.45	+1.93*	+1.37	+1.77*		
SANTA-F	+1.27	+1.03	+0.84	−0.83	−1.32	
SANTA	+1.48	+1.97**	+1.45	+1.83*	+0.78	+1.31
$h+63$ (<i>eff. N</i> ≈ 15)						
NLinear	+0.74					
Day-Token	+1.38	+0.05				
Per-Cell	+1.10	+1.43	+0.79			
SANTA-T	+1.34	+2.11**	+1.16	+2.11**		
SANTA-F	+1.26	+1.87*	+1.04	+1.15	−0.26	
SANTA	+1.37	+2.05**	+1.24	+1.78*	+0.87	+0.72

Table C.3: Pairwise HLN-corrected Diebold–Mariano statistics on MSE. Cell (r, c) is the statistic for row r against column c ; a positive value means row r has the lower MSE. NW HAC lag = $h-1$, two-sided. *** $p < .01$, ** $p < .05$, * $p < .10$.

	VAR	NLinear	Day-Token	Per-Cell	SANTA-T	SANTA-F
<i>h+10 (eff. $N \approx 100$)</i>						
NLinear	+3.14***					
Day-Token	+7.23***	+2.02**				
Per-Cell	+4.70***	+1.60	-1.38			
SANTA-T	+4.70***	+1.55	-1.33	+0.33		
SANTA-F	+4.56***	+1.32	-1.59	-0.63	-0.89	
SANTA	+5.20***	+1.49	-1.22	+0.41	+0.28	+1.63
<i>h+21 (eff. $N \approx 47$)</i>						
NLinear	+0.95					
Day-Token	+2.74***	+1.42				
Per-Cell	+1.88*	+1.72*	+0.06			
SANTA-T	+1.72*	+1.58	+0.18	+0.63		
SANTA-F	+1.82*	+1.68*	+0.16	+0.43	+0.17	
SANTA	+1.90*	+1.57	+0.14	+0.23	+0.21	+0.22
<i>h+42 (eff. $N \approx 23$)</i>						
NLinear	+0.29					
Day-Token	+1.44	+1.14				
Per-Cell	+1.24	+2.26**	-0.55			
SANTA-T	+1.29	+2.38**	-0.68	-0.83		
SANTA-F	+0.96	+1.27	-0.10	-0.68	-0.93	
SANTA	+1.47	+2.53**	-0.99	+1.54	+0.56	+1.24
<i>h+63 (eff. $N \approx 15$)</i>						
NLinear	-0.10					
Day-Token	+1.09	+0.87				
Per-Cell	+0.57	+1.51	-0.04			
SANTA-T	+0.95	+2.25**	-0.47	+2.03**		
SANTA-F	+0.74	+1.84*	-0.30	+1.00	-0.32	
SANTA	+1.05	+2.62***	-0.65	+1.37	+0.49	+0.52

Table C.4: Pairwise HLN-corrected Diebold–Mariano statistics on MAE. Cell (r, c) is the statistic for row r against column c ; a positive value means row r has the lower MAE. NW HAC lag = $h-1$, two-sided. *** $p < .01$, ** $p < .05$, * $p < .10$.

	VAR	NLinear	Day-Token	Per-Cell	SANTA-T	SANTA-F
<i>h+10 (eff. $N \approx 100$)</i>						
NLinear	-0.70					
Day-Token	+1.52	+1.54				
Per-Cell	-0.04	+0.95	-0.90			
SANTA-T	+1.25	+2.18**	+0.05	+2.06**		
SANTA-F	+0.95	+1.81*	-0.20	+1.63	-0.79	
SANTA	+1.08	+1.85*	-0.13	+1.77*	-0.44	+0.36
<i>h+21 (eff. $N \approx 47$)</i>						
NLinear	-0.89					
Day-Token	+0.25	+0.92				
Per-Cell	+0.18	+1.70*	-0.00			
SANTA-T	+1.91*	+3.00***	+1.44	+2.21**		
SANTA-F	+1.91*	+2.98***	+1.45	+2.35**	-0.22	
SANTA	+2.01**	+2.95***	+1.54	+2.36**	+0.07	+0.49
<i>h+42 (eff. $N \approx 23$)</i>						
NLinear	-0.68					
Day-Token	-1.19	-0.21				
Per-Cell	+0.81	+1.81*	+1.33			
SANTA-T	+1.86*	+2.56**	+2.14**	+1.75*		
SANTA-F	+1.90*	+2.90***	+2.11**	+2.13**	+0.05	
SANTA	+1.97**	+2.74***	+2.19**	+1.98**	+0.50	+0.62
<i>h+63 (eff. $N \approx 15$)</i>						
NLinear	-0.77					
Day-Token	-2.54**	-0.87				
Per-Cell	-0.27	+0.54	+1.34			
SANTA-T	+1.11	+1.57	+2.11**	+2.75***		
SANTA-F	+1.63	+2.27**	+2.35**	+3.08***	+1.03	
SANTA	+1.63	+2.11**	+2.37**	+3.07***	+1.45	+0.35

Table C.5: Pairwise HLN-corrected DM statistics on cross-sectional rank IC. Cell (r, c) is the statistic for row r against column c ; a positive value means row r has the higher IC. NW HAC lag = $h-1$, two-sided. *** $p < .01$, ** $p < .05$, * $p < .10$.

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